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IT496: Graduation Project Sprint 1 Report

Siyaq

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Glossary

The following Table 1 defines the traffic, legal and project-specific terms used in this report. Definitions taken from the Saudi Traffic Law and its Executive Regulations cite the relevant provision [4].

Term	Definition
Absher	The Ministry of Interior's e-services platform, which offers the Minor Accident Reporting service.
Accident case	The record of one accident in Siyaq: both parties' information, vehicles, descriptions, videos, analysis result and the expert's decision.
Accident investigator	The official who determines the fault percentage under the supervision of the competent officer (Regulation 7/60) [4]. In Siyaq, this role corresponds to the authorized expert.
Accident party	A driver involved in the accident. The first party creates the case and must upload one dashcam video; the second party joins through an invitation link and may upload one optional video.
Accident type	The scenario chosen by the first party when submitting a case: rear-end collision or lane-change accident.
Article 50	The article of the Traffic Law that contains the rules of the road: following distance, braking, lane changing, priority, overtaking and signalling [4].
Article 59	The article that defines traffic accidents and divides them into minor and serious accidents [4].
Article 60	The article that sets how responsibility and fault percentages in a traffic accident are determined [4].
Authorized expert	A qualified accident-assessment professional who reviews each case in Siyaq, may modify the suggested fault percentage or description, and approves the final result.
Basic fault ratio	In the Korean Fault Ratio Information Portal, the starting fault ratio assigned to a predefined accident scenario before adjustment factors are applied.
Case status	The stage of a case: Pending (waiting for the second party), In Progress (waiting for expert review) or Completed (approved).
Causal link	The direct or indirect connection between a driver's act and the accident, required for that driver to be held responsible (Regulation 1/60) [4].
Collision reconstruction	Recreating how a collision happened, for example as a map, timeline or 3D simulation; offered by Empiric Earth and Liablix, not by Siyaq.
Combine and consistency check	The step that places the events from both videos in one case record. If the videos disagree, the result is undetermined (U9).
Competent court	The court that decides responsibility when the authorities cannot determine it after completing the investigation (Regulation 6/60) [4].
Dashcam	A camera fixed in a vehicle that records the road. Siyaq uses front-facing and rear-facing cameras only.
Evidence frames	Video frames that show the detected events, attached to the report so the expert can see the evidence.
Executive Regulations	The detailed regulations issued under the Traffic Law. They contain the rules of the road and how accident responsibility is determined [4].
Failure to comply with the regulations	Doing something that violates the traffic regulations (Regulation 3/2/60) [4]. It is the ground recorded for every Siyaq rule.

Term	Definition
Fault percentage (fault ratio)	The share of responsibility assigned to each party. The Regulations define 100%, 75%, 50% and 25% (Regulation 2/60) [4]; Siyaq also uses 0% for a party with no fault.
First notice of loss (FNOL)	The first report sent to an insurer after an accident.
Following distance	The gap a driver must keep from the vehicle ahead in the same lane; it must increase as speed increases (Regulation 19/2/50) [4].
Force majeure	An external event beyond the driver's control. A driver who causes an accident due to force majeure is exempted from the public right (Regulation 4/60) [4].
Free-flow road	A main road that can only be entered or exited through dedicated, controlled entries and exits, and does not directly serve the properties beside it (Traffic Law, Article 2) [4].
Grounds of responsibility	The three reasons on which a fault percentage is based: negligence, lack of precaution and failure to comply with the regulations (Regulation 2/60) [4].
Intentional accident	A collision caused on purpose. It is not a traffic accident under the Law and is investigated by the police (Regulation 1/59) [4].
Invitation link	The link sent to the second party so they can join the same case.
Lack of precaution	Doing something wrong that shows poor foresight and failure to consider its consequences (Regulation 2/2/60) [4].
Lane line	A painted longitudinal line separating lanes. A dashed line may be crossed when safe; a continuous (solid) line must not be driven on or crossed (Regulation 9/7/6/2/6/50) [4].
Lane-change accident	A collision that happens while a vehicle moves from one lane into another; one of the two accident types supported by Siyaq.
Leading vehicle	The vehicle in front. It has priority, and its driver must not stop or change direction suddenly (Regulation 1/1/1/6/50) [4].
Mapping principle	Siyaq's rule for setting ratios: one party violated → 100% / 0%; both violated and one is the main cause → 75% / 25%; both contributed equally → 50% / 50%.
Minor (simple) accident	An accident that causes property damage and, at most, an injury that does not need hospital transfer (Regulation 2/59) [4]. Siyaq's scope is narrower: no injury at all, property damage only.
Najm assessor	The Najm employee who examines a minor accident, at the scene or remotely, and determines each party's fault percentage.
Najm for Insurance Services	The Saudi company that handles eligible traffic accidents for insured vehicles, including sending assessors and issuing accident reports.
Negligence	Failing to do something that caution required, where doing it would have prevented the accident (Regulation 1/2/60) [4].
Overtaking	Passing a vehicle travelling in the same direction. The overtaken vehicle must not speed up, and the overtaking vehicle must leave enough side distance (Regulations 4/2/6/50 and 5/6/2/6/50) [4].
Priority (right of way)	The right of one vehicle to proceed before another. For example, a vehicle going straight has priority over a vehicle changing lanes (Regulation 2/1/1/6/50) [4].
Property damage only	An accident with damage to vehicles or property and no injuries; the only kind of accident Siyaq supports.
Provision	A specific article, regulation item or schedule item of the Traffic Law and its Executive Regulations that a Siyaq rule refers to.

Term	Definition
Provision number	The number of a provision, read from right to left: the last number is the article and each number to its left is an item within the one to its right. For example, 2/1/1/6/50 is item 2 of item 1/1/6/50 under Article 50.
Public right	The state's right to penalize an offence, as opposed to the private right of the injured party to compensation.
Rear-end collision	A collision in which the front of one vehicle hits the rear of another vehicle in the same lane; one of the two accident types supported by Siyaq.
Recording vehicle	The vehicle whose dashcam recorded the submitted video; shown as vehicle A in the figures.
Remote damage assessment	A Najm service that assesses vehicle damage from submitted photos without attending the scene. It covers damage only, not fault.
Responsibility (liability)	A party's legal accountability for the accident. It arises when there is a causal link between the party's act and the accident (Regulation 1/60) [4].
Retention period	How long uploaded videos and case data are kept: about 90 days after the case is resolved, then deleted automatically.
Rule ID	The code of a Siyaq rule: R1 to R3 for rear-end collisions and L1 to L6 for lane-change accidents (Tables 9 and 10).
Rule-based fault estimation	The Siyaq component that applies the rules to the extracted evidence to produce a suggested fault percentage or an undetermined result.
Saudi Traffic Law	The Saudi law that regulates vehicles, drivers and road use, including accidents and violations [4].
Serious accident	An accident that causes death, an injury requiring hospital transfer, or major property damage (Regulation 3/59) [4]. Outside Siyaq's scope.
Service road	A road running beside a free-flow road that directly serves the properties along it (Traffic Law, Article 2) [4].
Siyaq	The name of the proposed system (Arabic سياق, "context"): a web-based decision-support system that analyzes dashcam video of a minor accident and suggests a fault percentage for expert review.
Siyaq rule	A predefined rule linking a situation and its evidence to the provision violated and a suggested ratio (Table 12).
Staged accident	An accident deliberately faked by a party; an offence under Article 63(2)(b) [4].
Sudden stop	Braking sharply without a traffic reason, such as an obstacle, queue or signal ahead. The Regulations prohibit it except when necessary or for safety (Regulation 5/2/50) [4].
Suggested ratio	The fault split proposed by a Siyaq rule. It is Siyaq's suggestion, not a ratio stated in the Regulations, and the authorized expert reviews it.
Supported scenarios	The two accident types Siyaq covers, rear-end collisions and lane-change accidents, recorded in clear weather and visibility conditions.
Traffic accident	An unintentional event involving at least one moving vehicle that causes human or property damage (Traffic Law, Article 2) [4].
Traffic event	A driving action extracted from the video, such as a lane change, a sudden stop, the turn-signal state or the following distance.
Traffic violation	An act that breaks a provision of the Traffic Law or its Executive Regulations, such as not signalling or not keeping enough distance.

Term	Definition
Turn signal	The indicator light a driver must switch on in good time before changing lanes or direction (Regulations 1/7/50 and 2/24/2/50) [4].
Undetermined result	The output when Siyaq cannot suggest a fault percentage, given with a reason code from U1 to U15 (Table 13). It is not a legal finding.

Table 1: Glossary of Terms

1 Introduction

Road traffic accidents are a significant global public-safety challenge, causing approximately 1.16 million deaths annually and substantial economic losses worldwide [1]. Traffic accident analysis may require examining multiple factors, including vehicle movements, driving behavior, traffic-rule compliance, and the sequence of events leading to the collision. When dashcam recordings are available, they can provide useful visual information about these events; however, extracting and interpreting the relevant information from video footage can require considerable manual effort.

Recent advances in Artificial Intelligence (AI) and computer vision provide an opportunity to support traffic-scene analysis by automatically detecting relevant objects, tracking their movements, and extracting information from video data [2] [3]. Rule-based reasoning can then be used to relate detected events and traffic violations to applicable traffic regulations. In Saudi Arabia, the Traffic Law and its Executive Regulations provide a proportional framework for determining fault percentages of 25%, 50%, 75% and 100% (Regulation 2/60), to which Siyaq adds 0% for a party with no assigned fault, based on factors including negligence, lack of precaution, and failure to comply with regulations [4]. Currently, this assessment relies on manual review by Najm assessors, leaving room for an automated and explainable alternative. Accident evidence can be submitted remotely through mobile assessment services [5], but it is for damage assessment only. Based on the solutions and research reviewed for this project, existing approaches address individual aspects of traffic accident analysis, such as automated evidence extraction, collision reconstruction, or responsibility assessment. However, no identified solution combines automated dashcam-video analysis, explicit rule-based fault-percentage estimation derived from relevant Saudi traffic regulations, supporting evidence and reasoning, and authorized expert review within the same workflow. This project proposes Siyaq, a web-based AI decision-support system for traffic accident analysis, where its main new contribution is estimation of fault percentages. The system allows an accident party to select one of the supported accident types and submit a case with a single dashcam video. The first accident party is required to provide a dashcam video, while the second accident party may optionally provide an additional dashcam video. Therefore, a case may include one or two dashcam videos. Each uploaded video is analyzed independently to detect relevant vehicles, movements, driving events, and traffic violations. The analysis results from the available videos are then combined into a single report, where rule-based logic derived from relevant Saudi traffic regulations is applied to generate a suggested fault percentage. The generated result, together with its supporting evidence and reasoning, is then presented to an authorized expert for review, modification if necessary, and final approval.

1.1 The Problem

Traffic accident analysis requires examining the sequence of events, vehicle movements, driving behavior, and applicable traffic rules in order to understand how an accident occurred and assess the responsibility of the involved parties. Currently, accident assessment relies heavily on manual review, where authorized personnel analyze the available evidence and interpret it according to traffic regulations. Although dashcam footage provides valuable visual information, extracting and interpreting the relevant evidence from one or more videos remains a time-consuming process that requires significant manual effort.



For example, in accidents involving a lane change or a failure to maintain a safe following distance, determining responsibility may require identifying which vehicle changed direction, which vehicle had priority, whether a traffic rule was violated, and how these events contributed to the collision. According to the Saudi Traffic Law and its Executive Regulations, accident responsibility is determined when a causal relationship exists between the action and the resulting accident (Regulation 1/60). Fault percentages are determined using proportional levels of 25%, 50%, 75% and 100% (Regulation 2/60), based on factors including negligence, lack of precaution, and failure to comply with traffic regulations [4]. There are many types of traffic accidents, and they range in severity where the least is property damage only. When a no-injury traffic accident happens where there is property damage only, the parties often wait at the accident location for an official from Najm or the traffic police to arrive for examination, fault determination and reporting. This happens even though the Executive Regulations already allow simpler handling: the vehicles in a simple accident may be moved once the parties have photographed them (Regulation 1/1/63), and a simple accident between insured vehicles may be documented without attending the scene (Regulation 17/61) [4]. The problem here is that the wait time can be long, causing delays and preventing accident parties from going to work or any other place when they are short in time. In addition, cars in the accident can obstruct the traffic flow during this wait time, causing more delays for other people on the street. Also, Traffic accident assessment requires authorized personnel to examine the available accident evidence and determine the responsibility of the involved parties. This process can require considerable time and manual effort, particularly when the available evidence must be reviewed and interpreted according to traffic regulations. Dashcam recordings can provide additional evidence about the sequence of events leading to an accident; however, extracting and interpreting relevant information from these recordings manually can also require significant effort. Therefore, there is a need for a decision-support system that assists authorized personnel by automatically analyzing available dashcam footage, identifying relevant traffic events and violations, and relating the extracted evidence to applicable traffic rules. The generated fault estimation serves as a recommendation supported by evidence, while the final decision remains the responsibility of the authorized expert.

1.2 The Solution

To address the identified problem, this project proposes Siyaq, a web-based AI decision-support system for traffic accident analysis, whose main new contribution is the estimation of fault percentages. The system allows an accident party to select the applicable accident type from the supported accident scenarios and submit a case. The first accident party is required to upload one dashcam video when submitting the case. The second accident party is then invited to join the same case and complete their required information, with the option to upload one additional dashcam video if available. Therefore, each case contains one mandatory dashcam video and may contain a second optional video. Each uploaded video is analyzed independently, and the generated results are combined into a single accident report.

Using computer vision techniques, Siyaq analyzes the available footage to detect and track relevant vehicles, examine their movements and driving behavior, and identify relevant traffic events and predefined traffic violations. Based on the accident type selected by the user and the information extracted from the footage, the system applies rule-based logic derived from relevant Saudi traffic regulations to generate a suggested fault percentage. The system uses the fault ratios of 25%, 50%,

75% and 100% defined in Regulation 2/60 of the Saudi Traffic Law and its Executive Regulations [4], and adds 0% for a party with no assigned fault.

The generated result is presented together with relevant supporting evidence and reasoning. An authorized expert reviews each accident case, including the accident information, supporting evidence, and suggested fault percentage, and may modify the suggested percentage if necessary before providing final approval. After expert approval, the accident parties can review the final approved result. The system is limited to no-injury traffic accidents involving property damage only, and the generated fault percentage serves as decision support for the authorized expert rather than as a legally binding decision.

To protect user privacy, privacy safeguards are applied to the final report presented to accident parties and expert users, rather than during internal video processing. Before submitting a case, users must provide explicit consent confirming their agreement to the use of the uploaded dashcam footage for accident analysis. Uploaded videos and associated data will be retained for approximately 90 days following case resolution, after which they will be automatically deleted.

Through this approach, Siyaq provides a structured and explainable process that combines computer vision, rule-based reasoning, fault-percentage generation, and human expert review within a single web-based system. This helps streamline accident assessment, reduce manual effort, support faster processing, and enable more efficient decision-making while preserving human oversight.

1.3 Product

1.3.1 Product Vision

The product vision statement for Siyaq is as follows:

سِياق
SIYAQ

Product vision

Siyaq · web-based AI decision support for traffic accident analysis

For	**parties involved in traffic accidents and authorized reviewers** *(a target customer)*
Who	**need a clear and accessible way to submit available dashcam evidence and review a suggested fault-percentage assessment** *(statement of need or opportunity)*
The	**Siyaq** (product name) is a **web-based platform** *(product category)*
That	**provides an AI-powered decision-support system for traffic accident analysis and suggested fault-percentage generation** *(key benefit / reason to buy)*
Unlike	**Najm's current manual assessment process, which relies on a human assessor to determine fault** *(primary competitive alternative)*
Our product	**provides an automated system that combines video analysis, fault-percentage generation, explainable rule-based reasoning, and expert review within a single platform** *(statement of primary differentiation)*

1.3.2 Product Roadmap

The following figure shows the product roadmap for Siyaq.

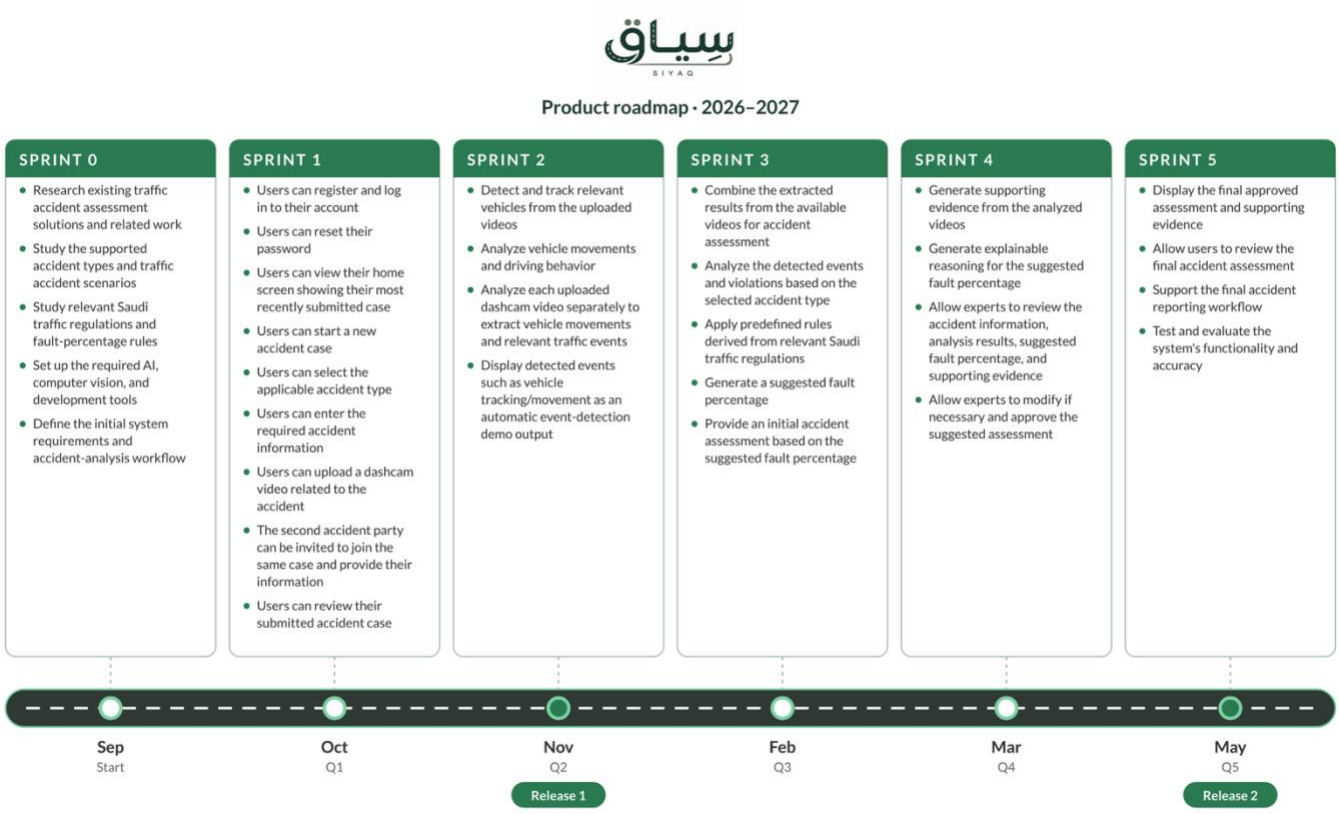


Figure 1: Siyaq product roadmap.

1.3.3 Objectives

Product Objectives (customer focus-value):

- The first accident party is required to submit one dashcam video, while the second accident party may optionally submit one additional video related to the same accident. Each uploaded video is analyzed independently.
- Users can select the applicable accident type from the supported accident categories when submitting a case.
- Users can view relevant detected traffic events, vehicle movements, and traffic violations related to the accident.
- Users can receive a suggested fault percentage based on the analyzed evidence and predefined rules derived from relevant Saudi traffic regulations.
- Users can view clear supporting evidence and reasoning for the suggested fault percentage.
- Experts can review accident information, supporting evidence, and the suggested fault percentage, modify it if necessary, and provide final approval.

Project Objectives (solution focus-plan):

- Conduct domain research to understand traffic accident analysis and identify the requirements relevant to the supported accident types.
- Identify, acquire, and prepare suitable dashcam video datasets for the supported accident types.
- Study relevant Saudi traffic regulations and define the traffic rules and rule-based logic required for fault-percentage generation.
- Define the functional, non-functional, privacy, and security requirements for accident parties and expert users.
- Develop and evaluate computer vision components for vehicle detection, tracking, and movement analysis.
- Develop methods for identifying relevant traffic events and predefined traffic violations and generating supporting evidence from dashcam videos.
- Develop a rule-based component that uses the extracted evidence and relevant Saudi traffic rules to generate a suggested fault percentage.
- Develop a web-based application with interfaces for accident parties and expert users, including case submission, result review, and expert approval.
- Design and implement the required database and server-side components for managing accident cases, analysis results, and expert decisions.
- Integrate the computer vision, rule-based reasoning, expert review workflow, database, and web components into a complete system.
- Evaluate the overall system using appropriate performance metrics and relevant baselines, including its computational performance and video-processing requirements.

Learning Objectives (student focus):

- Learn advanced computer vision techniques for analyzing dashcam videos.
- Learn how to apply object detection and multi-object tracking techniques to identify and follow vehicles in traffic scenes.
- Learn how to process, prepare, and evaluate video datasets for computer vision applications.
- Learn techniques for analyzing vehicle movements and identifying relevant traffic events and predefined traffic violations from video footage.
- Learn how to design and implement rule-based reasoning that combines AI-extracted evidence with predefined traffic rules to generate explainable results.
- Learn how to evaluate computer vision and AI components using appropriate performance metrics and analyze their limitations and edge cases.
- Learn privacy-preserving and secure data-handling techniques for sensitive accident video and user information.



1.3.4 Scope

This project proposes Siyaq, which is a web-based AI decision-support system for traffic accident analysis and estimation of fault percentages. Due to the huge domain and large number of cases that can exist, the project will have some limitations to be suitable for the time of a graduation project. This project focuses on analyzing one or two dashcam videos related to the same minor traffic accident. The first accident party is required to provide one dashcam video, while the second accident party may optionally provide one additional video if available. The supported accidents are limited to minor, no-injury traffic accidents involving property damage only and recorded in clear weather and visibility conditions This scope is narrower than the legal definition of a simple accident, which may include a minor injury that does not require hospital transport (Regulation 2/59); accidents with serious damage to property are classed as serious accidents (Regulation 3/3/59) and are also excluded [4]. The system is limited to two accident scenarios: rear-end collisions and lane-change accidents, which were identified during expert consultation as representing approximately 80% of traffic accidents in Saudi Arabia. The accident party selects the applicable accident type from the supported scenarios when submitting the case.

The proposed solution will be developed as a web-based application in Arabic. It analyzes one or two dashcam videos related to the same accident, one per accident party, to generate a suggested fault percentage, together with explainable reasoning and supporting evidence based on predefined rules derived from relevant Saudi traffic regulations.

The system supports two types of users: accident parties and authorized expert users. Accident parties can submit accident cases and review the final approved results, while expert users review the generated analysis and approve or modify the suggested fault percentage before issuing the final decision.

The system is intended to support traffic accident analysis and human decision-making and will not replace human judgment or make legally binding decisions. Accident scenarios beyond the two specified types, synchronization between multiple dashcam recordings, low-visibility conditions, adverse weather conditions, and accidents involving injuries or major property damage are outside the scope of this project.



1.3.5 Hardware/Software Tools and Cost

Hardware Tools	
Name and Description	**Cost**
Personal Laptops - Used for system development, programming, testing, and running the web application.	0 SAR - Already available
GPU-Enabled Computing Environment - Used for training and/or running the computer vision models on dashcam videos. The exact GPU specification will be selected based on the model requirements and available computing resources.	0-500 SAR (using free resources when available or a cloud GPU subscription such as Google Colab Pro).
Application Server / Cloud Server - Used to host the web application, backend services, and support the accident-analysis workflow.	0-500 SAR (depending on the selected hosting platform and whether free-tier services are sufficient).
Software Tools	
Name and Description	**Cost**
Python – Programming language used for backend development and implementing computer vision components.	Free
Visual Studio Code – IDE used for writing, testing, and managing project code.	Free
OpenCV – Library for video processing and frame analysis.	Free / Open source
YOLO – Object detection model for detecting vehicles in dashcam videos.	Free / Open source (depending on model license)
ByteTrack – Multi-object tracking method for following detected vehicles across video frames.	Free / Open source
GitHub – Version control and source code management.	Free
Flask – Python web framework used to develop the backend and connect the web application with other system components.	Free / Open source
Neon PostgreSQL – Cloud database used to store user information, accident cases, and related system data.	Free tier available
Cloudinary API – Cloud storage service used to upload, store, and retrieve accident videos.	Free tier available; paid plans available
Brevo API – Email delivery service used to send OTP verification codes and invitations to accident parties.	Free tier available
OpenStreetMap Tile API – Provides map tiles for displaying accident locations on an interactive map.	Free public service (subject to usage policy)
Nominatim API – Reverse geocoding service used to convert selected accident coordinates into readable Arabic location names.	Free public service (subject to usage policy)
Leaflet – JavaScript library used to display interactive maps and allow users to select accident locations.	Free / Open source



Table 2: Software and Hardware Tools requirements.

1.4 Scrum Team

1.4.1 Skill Set Requirements

Technical Skill Required	What is the current level of the team (*beginner- intermediate- advanced*) for each skill? How will the gap be bridged? *(if necessary)* Learning plan
Computer Vision (object detection and tracking)	Beginner. Will strengthen through relevant documentation, tutorials, and practical experimentation with dashcam videos.
Video Processing and Dataset Preparation	Beginner. Will develop through practical work on preparing, processing, and evaluating video datasets.
AI/Computer Vision Evaluation	Beginner. Will learn and apply appropriate evaluation metrics for detection, tracking, and violation identification.
Rule-Based Decision-Support Systems	Beginner. Will study rule-based reasoning and its application to detect traffic events and predefined traffic rules.
Backend development	Advanced. Builds on prior coursework and will be extended to integrate the AI and rule-based components with the web application.
Frontend development	Advanced. Builds on prior coursework and course projects.
Data Storage/Database Design	Intermediate. Builds on prior coursework and will be applied to system data and analysis results.
Data privacy & security practices	Beginner. Will study and apply appropriate privacy and security practices for handling dashcam videos and system data.

Table 3: Technical Skills Requirements.

1.4.1.1 Learning

To achieve the required learning outcomes for this project, the team has begun exploring the technologies, datasets, and analysis techniques relevant to dashcam-based traffic accident analysis. Through the reviewed research, the team has developed an initial understanding of how video data can be processed across consecutive frames to extract information about vehicle movements and traffic events. The team has also become familiar with object-detection approaches such as YOLO, which can be used to identify relevant vehicles in video frames, as well as tracking and temporal analysis techniques for examining vehicle movement over time.

In parallel, the team has been reviewing research related to accident analysis and fault assessment to understand how extracted video evidence can be used in later stages of the system. This includes studying approaches that detect lane-changing behavior from temporal lane information and research that combines YOLO-based visual detection with rule-based inference to determine accident responsibility. The team has also reviewed relevant traffic-accident and driving datasets to understand the types of video data available for future development and evaluation.

Additionally, the team has been studying rule-based and regulation-grounded approaches for fault assessment. The reviewed research helped the team understand how visual evidence can be separated

from the reasoning stage, where detected movements and traffic events are evaluated using explicit traffic rules. At the current stage, this learning is primarily research-based and is being used to guide the selection of suitable tools, datasets, and techniques before practical implementation begins in the upcoming sprints.

1.4.2 Roles and Responsibilities

Scrum Team	
Product Owner:	Dr. Hessah Alsaaran
Developers:	Raneem Aloraini Lama Almubarak Jana Almengash Waad Alghamdi
Scrum Master (SM):	Dr. Hessah Alsaaran
Stakeholders:	Graduation Project Examination Committee.

Table 4: Scrum Team Table.

2 Background

Traffic accident assessment requires understanding how an accident occurred, what actions were taken by the involved drivers, and how these actions relate to traffic regulations. In Saudi Arabia, minor traffic accidents are generally handled through the traffic authorities or Najm for Insurance Services, depending on the circumstances of the accident. The assessment process involves collecting information about the accident and its parties, examining the accident scene, and determining the responsibility of each party.

Siyaq focuses on supporting this process for minor traffic accidents by using dashcam video as an additional source of evidence. Computer vision is used to extract information about vehicle movements from the video, while predefined traffic rules are used to generate a suggested fault percentage. The following sections explain the existing accident-handling process in Saudi Arabia, the rules used to determine responsibility, the accident types supported by Siyaq, the role of dashcam footage, and the technologies used by the proposed system.

2.1 Traffic Accident Assessment in Saudi Arabia

Traffic accident assessment in Saudi Arabia involves reporting the accident, documenting the required information, and determining the responsibility of the involved parties according to the applicable traffic regulations. For minor accidents, electronic services are also available to simplify the reporting process. This process and the information collected during it provide the basis for understanding how Siyaq can support accident analysis using additional video evidence.

2.1.1 Current Traffic Accident Process

When a traffic accident occurs, the involved parties must report it through the appropriate accident-reporting channel. The Ministry of Interior provides a Minor Accident Reporting service through the

Absher platform. According to the official National Platform [18], the service allows a beneficiary to report a minor traffic accident electronically by completing an accident registration form and entering the required accident details

For insured vehicles, Najm for Insurance Services also provides services for reporting and handling eligible traffic accidents. Depending on the circumstances of the accident, the process may involve reporting the accident, identifying its location, documenting the accident and the involved vehicles, and completing the required assessment procedures.

After the necessary information has been collected and the accident has been assessed, the parties can proceed with the remaining accident and insurance procedures. Siyaq does not replace the official reporting or assessment process. Instead, it aims to provide additional evidence that can support the analysis of how the accident occurred.

2.1.2 Accident Information Collection

Accident assessment requires sufficient information and evidence to document the circumstances of the collision. According to the Ministry of Interior [18], each party reporting a minor accident is required to document the incident by taking photographs that can be uploaded through the electronic service. The required photographs include the point of impact, accident markings, accident location, and the vehicle from the front, rear, right, and left sides. Visible evidence at the scene, such as skid marks, fluids, or stains on the ground, should also be documented.

This information provides important evidence about the accident scene and the condition of the vehicles after the collision. However, photographs mainly capture specific moments after the accident and may not fully show the sequence of events that led to the collision. For example, they may show the final position of a vehicle but not whether it changed lanes or how the distance between two vehicles changed before impact.

Siyaq therefore uses dashcam footage as an additional source of evidence. Video provides a continuous sequence of events that can show vehicle positions and movements before and during the collision, complementing the information collected through photographs.

2.1.3 Time and Consequences of the Current Process

The time required to handle a traffic accident can vary depending on the accident location, traffic conditions, and the procedures required to document and assess the case. A study of the existing Najm accident-handling process [19] estimated an approximate total cycle time of 1 hour and 28 minutes**.** The flow analysis includes activities such as reporting the accident, assigning a Najm worker, verifying the accident location, traveling to the accident scene, verifying the drivers' identities, taking photographs, assessing the accident, and completing the required assessment procedures. The analysis estimates approximately 20 minutes for reaching the accident location, 8 minutes for verifying the drivers' identities, and 10 minutes for taking photographs [19].

The reported 1 hour and 28 minutes represents the approximate total cycle time of the existing Najm process analyzed in the study rather than an official average for every accident handled by Najm. The actual duration may vary depending on the circumstances of each accident. The study also identifies delays in reaching the accident location and manual procedures as factors that may increase the time required to complete the process [19].

In addition to the time required to complete the accident procedures, delays in handling accidents can contribute to traffic congestion and affect other road users [19]. The process also requires resources for traveling to the accident scene, documenting the accident, and completing the assessment.

Siyaq aims to support this process by using dashcam footage and digital accident information as additional evidence. This provides information about the events before and during the collision that can support the assessment process while maintaining expert involvement in the final decision.

2.1.4 Saudi Traffic Rules for Fault Determination

In Saudi Arabia, determining responsibility for a traffic accident requires examining the circumstances of the accident and the driving actions that contributed to its occurrence. The assessment is related to whether the involved drivers followed Saudi traffic regulations and whether a driver's action contributed to the collision. Saudi traffic-safety data identifies behaviors such as failing to maintain a safe distance, sudden lane deviation, violating traffic regulations, distracted driving, and driving in the wrong direction as causes of traffic accidents [20].

Under the Executive Regulations, responsibility for a traffic accident arises when there is a direct or indirect causal link between a driver's act and the accident (Regulation 1/60). The fault ratio is set at 100%, 75%, 50% or 25% on three grounds (Regulation 2/60): negligence, which is failing to take an action that caution requires and that would have prevented the accident (Regulation 1/2/60); lack of precaution, which is a wrong act that shows a lack of foresight and of the consequences (Regulation 2/2/60); and failure to comply with the regulations (Regulation 3/2/60). Responsibility is described by the act that was committed or omitted (Regulation 3/60), and the ratio is determined by the accident investigator under the supervision of the competent officer (Regulation 7/60) [4].

Therefore, determining responsibility requires considering the sequence of events leading to the accident rather than relying only on the final position of the vehicles or the point of impact. The actions of the involved drivers before the collision provide important information for understanding how the accident occurred, and which traffic rules may have been violated.

This approach is relevant to Siyaq because the system analyzes observable vehicle actions from dashcam footage and evaluates the detected events using predefined rules based on Saudi traffic regulations. The system therefore considers evidence from the events leading to the collision before generating a suggested fault assessment.

Siyaq represents the suggested responsibility using 0%, 25%, 50%, 75%, and 100%. A value of 0% represents no assigned fault for a party, while the remaining percentages represent different levels of responsibility. The generated percentage is a suggested assessment based on the available evidence and applicable rules and is reviewed by an authorized expert before final approval.

2.2 Accident Types Supported by Siyaq

Different accident types involve different vehicle movements and require different evidence when assessing responsibility. Siyaq currently supports two accident scenarios: rear-end collisions and lane-change accidents. For each accident type, the system examines specific visual evidence from the submitted dashcam footage before applying the relevant predefined rules.

2.2.1 Rear-End Collisions

A rear-end collision occurs when the front of one vehicle collides with the rear of another vehicle. Although the collision position provides useful information, it does not fully explain the sequence of events that caused the accident.

In Siyaq, dashcam footage is analyzed to identify relevant information about the vehicles and their actions before and during the collision. This information provides additional context about how the rear-end collision occurred and is used as evidence in the accident analysis.

Rear-end collisions are not automatically treated as 0%-100% fault cases in Siyaq. The available evidence is analyzed to determine how the actions of the involved vehicles contributed to the accident. The identified events are then evaluated using predefined rules to generate a suggested fault percentage.

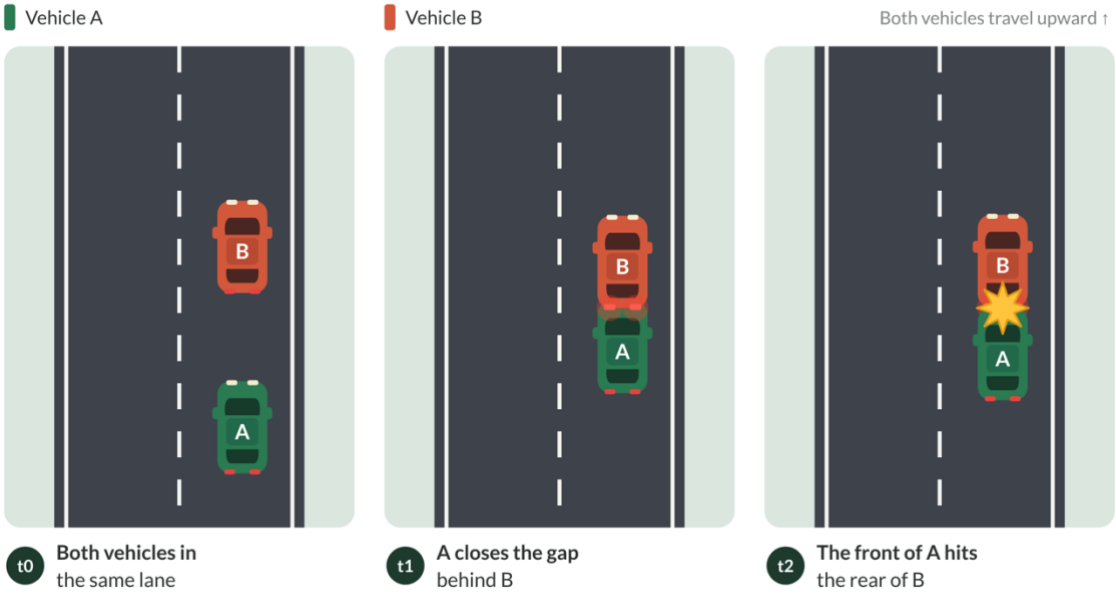


Figure 2: Rear-end collision scenario.

2.2.2 Lane-Change Accidents

A lane-change accident occurs when a collision is associated with a vehicle moving from one traffic lane into another. The actions of the involved vehicles before the collision are particularly important when analyzing this type of accident.

In Siyaq, dashcam footage is analyzed to identify relevant information about the vehicles and the events associated with the lane change. This information provides additional context about the sequence of events leading to the collision and is used as evidence in the accident analysis.

A lane-change accident does not automatically result in a fixed fault percentage. The available evidence is considered before predefined rules are applied to generate a suggested fault distribution.

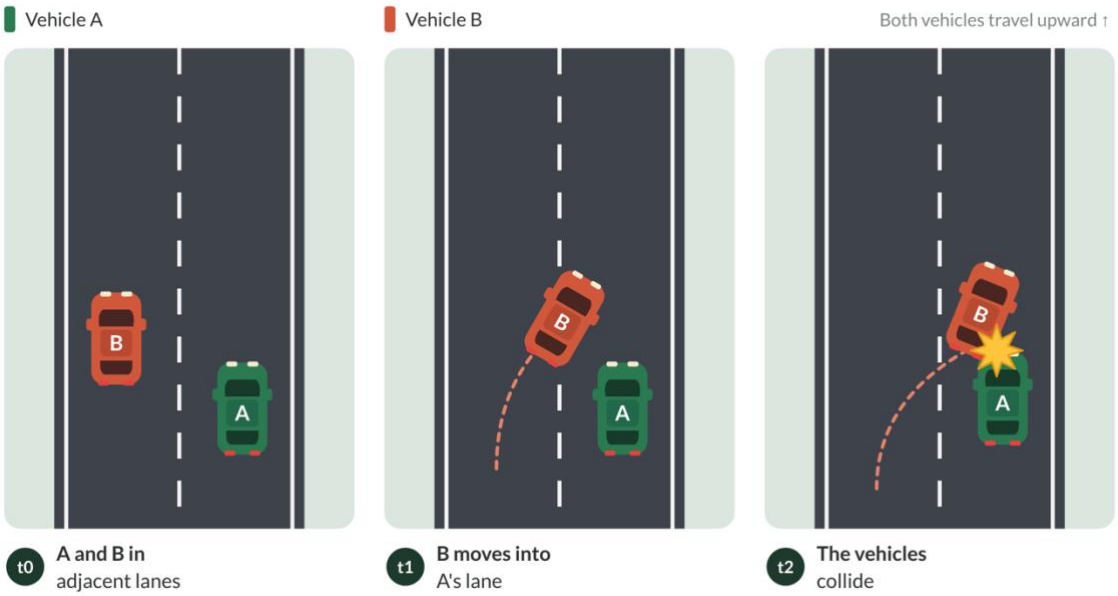


Figure 3: Lane-change accident scenario.

2.3 Dashcam Video as Accident Evidence

Dashcam recordings provide visual information about events occurring before and during a traffic accident. Unlike photographs taken after a collision, video contains a sequence of frames that can show changes in vehicle positions and movements over time. However, the evidence available in a recording depends on the direction of the camera and what is visible within its field of view.

2.3.1 Dashcam Camera Types

Dashcam systems may contain cameras facing different directions. A front-facing camera records the road ahead of the vehicle, while a rear-facing camera records the area behind it. Some dashcam systems also include an interior-facing camera that records the inside of the vehicle.

For Siyaq, the relevant camera types are front-facing and rear-facing road cameras because they provide visual information about the road environment and surrounding vehicles. Interior-facing footage is not used because it primarily records the vehicle cabin rather than the road evidence required for the supported accident scenarios.

The evidence available from a road-facing camera also depends on its field of view. Therefore, a vehicle or maneuver that occurs outside the visible area of the recording may not be captured and cannot be directly analyzed by the system.

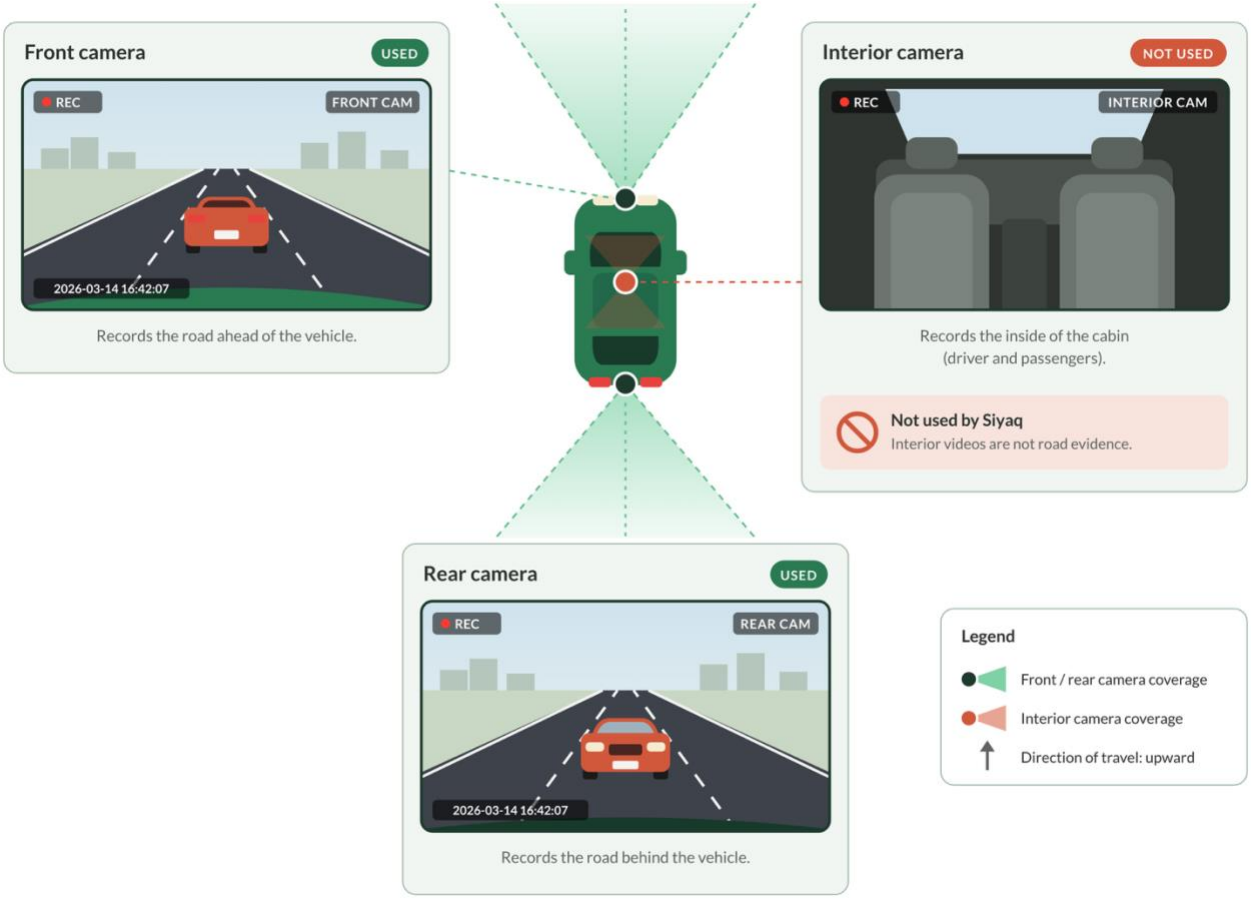


Figure 4: Dashcam camera types and their relevance to Siyaq.

2.3.2 Video Selection Based on Accident Type

The first accident party is required to upload one relevant dashcam video to Siyaq, while the second accident party may optionally upload one additional relevant video if available. The appropriate recording depends on the accident type and the position of the other vehicle.

For a rear-end collision, if the other vehicle is located in front of the accident party's vehicle, the front-camera recording is more appropriate because it may show the leading vehicle, the distance between the vehicles, and their movements before the collision. If the party's vehicle is struck from behind, the rear-camera recording is more appropriate because it may show the approaching vehicle and its movement before impact.

For a lane-change accident, the selected recording should be the road-facing video that provides the clearest view of the other vehicle and the lane-changing maneuver. Therefore, the appropriate camera depends on the relative positions of the vehicles and the available visual evidence rather than requiring the same camera direction for every accident.

2.3.3 Single-Camera Availability

Having both front and rear cameras is not required to use Siyaq. If an accident party has only one road-facing dashcam, the available recording can be submitted.

However, the system can analyze only the evidence visible in that recording. If an important vehicle, lane boundary, or maneuver occurs outside the camera's field of view, the footage may not contain sufficient information to evaluate that event.

Siyaq therefore does not assume that an event occurred when it cannot be observed in the submitted video. Each accident party uploads a maximum of one relevant recording. As a result, an accident case may contain one video when footage is available from only one party or two videos when both parties provide relevant recordings. Each submitted video is analyzed independently as a source of evidence.

2.4 Computer Vision for Traffic Video Analysis

Computer vision allows information to be automatically extracted from images and videos. In Siyaq, computer vision is used to process dashcam recordings and identify observable information about vehicles and their movements. The analysis includes video and frame processing, vehicle detection, and the examination of vehicle movement across consecutive frames.

2.4.1 Video and Frame Processing Using OpenCV

OpenCV is an open-source computer vision library that provides tools for processing images and videos. It can be used to read video recordings, access individual frames, and prepare visual data for further analysis.

In Siyaq, OpenCV is used to process uploaded dashcam recordings and access their individual video frames. Processing the recording as a sequence of frames allows the system to examine visual information at different moments before and during the collision.

2.4.2 Vehicle Detection Using YOLO

YOLO (You Only Look Once) is an object-detection model that can identify and locate objects within images and video frames. It detects objects such as vehicles and represents their locations within the frame.

In Siyaq, YOLO is used to detect relevant vehicles within individual dashcam frames. The detected vehicles can then be examined across consecutive frames to support the analysis of their movement.

2.4.3 Vehicle Tracking Using ByteTrack

ByteTrack is a multi-object tracking method that links the detections produced in each frame into continuous tracks, so that each vehicle keeps the same identity across consecutive frames [6]. Unlike trackers that keep only high-confidence detections, ByteTrack also associates low-confidence detections with existing tracks, which helps when a vehicle is partly occluded or blurred. In Siyaq, ByteTrack follows the vehicles detected by YOLO so that their paths, and the gap between them, can be analyzed over time.

2.4.4 Vehicle Movement and Traffic-Event Analysis

After the relevant vehicles are detected, their positions across consecutive frames are examined to identify movements and traffic events. The information required from this analysis depends on the accident scenario.

For rear-end collisions, the analysis focuses on the relative positions of the vehicles, their movement, and changes in the distance between them before the collision. For lane-change accidents, the analysis



focuses on lateral vehicle movement, visible lane boundaries, and the sequence of vehicle positions before the collision.

The purpose of the computer vision component is to identify observable evidence rather than determine fault directly. The extracted evidence is then provided to the rule-based component, where it is interpreted according to the predefined rules associated with the accident scenario.

2.5 Fault Estimation and Decision Support

Visual evidence alone does not determine the responsibility of the accident parties. The identified vehicle movements and events must be interpreted according to predefined traffic rules before a fault percentage can be suggested. Siyaq therefore separates the extraction of visual evidence from the process of estimating fault.

2.5.1 Rule-Based Fault Estimation

Siyaq uses rule-based reasoning to connect the evidence identified in the dashcam footage with predefined traffic rules. Once the relevant vehicle movements and events have been extracted, they are evaluated according to the supported accident scenario.

For example, detecting that a vehicle moved across a lane boundary shortly before a collision provides evidence of a lane-changing maneuver. This observation alone is not treated as the final fault decision. Instead, it is evaluated together with other available evidence, such as the positions and movements of the involved vehicles.

Based on the combination of evidence and predefined rules, Siyaq generates a suggested fault percentage of 0%, 25%, 50%, 75%, or 100%. In the system, 0% represents no assigned fault, while the remaining values represent different levels of responsibility.

The system also provides the evidence and reasoning associated with the suggested percentage. This allows the result to be understood rather than presenting a fault percentage without an explanation.

2.5.2 Explainability and Expert Review

Providing an explanation for a suggested fault percentage is important because the percentage alone does not show which evidence influenced the result. Siyaq therefore presents the relevant evidence and reasoning together with the suggested fault percentage.

An authorized expert reviews the accident information, the available dashcam evidence, and the system's suggested fault percentage. The expert can modify the suggested percentage when necessary before approving the final assessment.

This creates a human-in-the-loop approach in which computer vision and rule-based reasoning assist the assessment process without independently making the final decision. Siyaq therefore functions as an AI-based decision-support system rather than an independent system for making legally binding fault decisions.

2.6 Supporting Tools

Several technologies support the development and implementation of Siyaq. OpenCV is used for video and frame processing, YOLO for detecting vehicles in submitted accident recordings, and ByteTrack for tracking detected vehicles across frames.



The web application is developed using Flask, a Python web framework, and deployed on Vercel. Neon PostgreSQL serves as the cloud database for storing user information, accident reports, and case details. Cloudinary API is used to store uploaded accident videos and enable their retrieval when needed.

Additionally, Brevo API is used to send email verification codes (OTPs) and invitations to the other party involved in an accident, allowing both parties to participate in the same case. OpenStreetMap is used to display accident locations through an interactive map, while its Nominatim API converts selected coordinates into readable Arabic location names. Leaflet is used to display and manage the interactive map within the web application.

3 Literature Review

This chapter reviews the research and existing solutions related to traffic accident analysis and fault assessment. It first discusses previous studies that use dashcam video and computer vision to understand accident events, extract relevant evidence, and analyze vehicle movements. It then reviews approaches that use explicit traffic rules and regulatory knowledge to assess accident fault and explains how these approaches relate to the design of Siyaq.

The chapter also identifies the research gap based on the reviewed work and positions Siyaq in relation to existing research. Finally, a competitive product analysis compares Siyaq with existing accident assessment and analysis solutions in terms of their main features, strengths, and limitations. Together, these sections establish the research and practical context for the proposed system.

3.1 Related Work

Traffic accident analysis requires understanding the sequence of events preceding a collision, the movements and behavior of the involved vehicles, compliance with traffic regulations, and the relationship between these factors and the resulting accident. Dashcam recordings can provide valuable visual evidence for such analysis; however, extracting the relevant information from video footage and interpreting it according to traffic regulations requires considerable manual effort. Siyaq addresses this challenge through a combination of computer vision and rule-based reasoning: the system analyzes one or two dashcam videos to detect and track relevant vehicles, examine their movements, identify predefined traffic events and violations, and use the extracted evidence to generate a suggested fault percentage based on relevant Saudi traffic regulations.

The reviewed work can be organized along three complementary lines: understanding accidents from dashcam video, extracting the specific information required for fault assessment, and reasoning about fault itself. Earlier studies concentrated mainly on detecting or anticipating accidents, while more recent work addresses the harder task of determining fault from accident evidence. The subsections below follow this progression and, for each theme, identify what the work established, how the approaches differ, what limitations were reported, and what those findings imply for the design of Siyaq.

Accident Understanding from Dashcam Video

Accident anticipation research established that computer-vision models can extract meaningful spatial and temporal information from dashcam video. DRIVE [3] is a deep reinforcement learning approach

that anticipates traffic accidents while producing a visual explanation of its predictions. The method simulates both bottom-up and top-down visual attention in a dashcam observation environment, so that the decision of the agent can be explained through the attended regions, and it was evaluated on multiple real-world traffic accident datasets. DRIVE [3] predicts an accident before it occurs rather than analyzing fault afterwards, so it is not a direct competitor to Siyaq. It is nevertheless relevant to two of Siyaq's design decisions. First, traffic events have to be analyzed across a sequence of frames rather than from isolated images, because what matters is how the situation develops. Second, in safety-related applications an automated decision becomes considerably more useful when it is presented together with the visual evidence that produced it, which is the same principle Siyaq applies when displaying the detected events supporting a suggested fault percentage. These capabilities, however, concern understanding what is happening in the video; the work discussed next takes the further step of asking what that footage can say about fault.

Extracting Evidence for Fault Assessment

A second line of work moved from detecting accidents toward extracting information that is specifically useful for fault assessment. Lee and Lee [7] investigated split liability assessment directly from dashcam footage using three-dimensional convolutional neural networks (3D CNNs). Instead of treating accident analysis as a short-clip recognition problem, their method aggregates information predicted across longer video sequences and classifies accident cases according to the accident types used to determine the degree of negligence.

This work established two points that matter for Siyaq. It showed that dashcam video can support more than the recognition that an accident occurred, and it showed that the temporally relevant part of the recording must be preserved rather than reduced to a single representative frame. The approach differs from Siyaq in how the conclusion is produced: the relationship between the video and the assigned negligence is learned from labeled examples, so the reasoning is implicit in the model. Siyaq instead separates evidence extraction from an explicit reasoning stage grounded in written traffic regulations, so that the applicable rule can be presented to the reviewer alongside the result. Neither choice is inherently better; the explicit form is preferred here because an authorized expert must be able to verify and modify the outcome.

The evidence a fault assessment depends on is also scenario-specific, and one of the two scenarios supported by Siyaq is the lane-change accident. Wang et al. [8] addressed the automated detection of lane-change events in naturalistic driving video, where the practical difficulty is that lane markers appear as weak patterns only a couple of pixels wide. Their method accumulates lane-marker candidates over time instead of deciding from individual frames, which makes detection robust to low resolution and to various interferences. Tested on 490 hours of naturalistic driving video from 63 drivers, it reached a sensitivity of 94.8% on a manually coded 10-hour subset, and the events detected in the larger set were still reviewed manually and classified as true or false lane changes. Two points follow for this project. The study confirms that a lane change can be recovered from temporal visual information in ordinary driving video, which supports treating lane-change identification in Siyaq as an evidence-extraction step rather than as information the user is asked to supply. At the same time, the study was designed for data reduction in naturalistic-driving research rather than for accident-fault assessment, it detects the lane changes of the recording vehicle from lane-marker information, and its

output still required manual verification; it therefore does not by itself establish that every relevant manoeuvre of a surrounding vehicle can be detected, and Siyaq is not expected to reproduce this particular algorithm. Where the manoeuvre of another involved vehicle matters, vehicle tracking, and movement analysis may therefore be required in addition to lane-marker information. In both cases the detected lane change remains intermediate evidence: it describes what happened and relating it to fault is the task of the rule stage discussed next.

A further question concerns what happens when more than one recording of the same accident is available, since Siyaq may receive one or two dashcam videos. Darms et al. [9] describe the fusion strategies used in driver assistance systems, distinguishing fusion at the raw-data, feature and decision levels, and separating low-level processing such as tracking and object discrimination from high-level processing such as situation assessment; they note that these abstraction layers allow more flexible and modular systems to be built. Their work concerns sensor and data fusion within a driver assistance system rather than two independent dashcam recordings submitted by different accident participants, and it does not address accident-fault assessment, or the architecture proposed here, so it is used only as conceptual support for separating the processing of individual information sources from any later integration of their results. On that basis, Siyaq does not perform raw-video or synchronized multi-camera fusion; instead, each recording is processed independently, and the resulting higher-level evidence may be considered together during expert review. If the independently extracted evidence is inconsistent or insufficient, the discrepancy can instead be presented for authorized expert review rather than being resolved through an assumed automatic fusion rule.

Rule-Based Fault Assessment

Where the approaches above stop at producing evidence, a third line of work makes the rules that turn evidence into fault explicit. The work most closely related to Siyaq in this respect is that of Yawovi et al. [10], who proposed an object-detection-based responsibility assessment system for crossroad vehicle collisions. Their system takes the video recorded by the driving recorder of one of the involved vehicles as input and consists of three modules: crash-time detection, traffic-sign detection, and responsibility assessment. YOLOv5 is used for visual detection, trained on a purpose-built dataset of 1,530 head-on crash images extracted from online videos, while an inference engine applies Japanese road rules to derive the degree of negligence of each party. The authors state that the rule-based component was chosen so that users can understand the reasons behind the assessment. The significance of this study is that it establishes the feasibility of the pipeline that Siyaq also adopts evidence derived from video is passed to an explicit set of traffic rules, which in turn produces a responsibility assessment that can be explained to the user. It is equally instructive in what it reports as limitations. The system was evaluated on 180 crossroad crash videos, and its performance depended strongly on lighting conditions, the visibility of the collided object, and the viewing distance to the traffic light, with accuracy dropping sharply under poor visibility. The reported version handles crossroad crashes with traffic lights involving two cars, processes a single driving-recorder video at a time, and requires the user to enter several factors manually, including road width, vehicle speed, vehicle direction and accident location. Although crossroad collisions fall outside the initial scope defined for Siyaq, the pipeline demonstrated here — video-derived evidence passed to an explicit rule set — is the one Siyaq adapts to its own supported scenarios.

The same group later extended the approach using open data [11]. The extended system covers six types of head-on and angle crashes occurring without traffic lights and six adverse weather conditions, and it reports improved performance, particularly at night without traffic lights. This follow-up work is informative in two ways. It confirms that the rule-based pipeline can be broadened beyond the initial configuration, and it also shows that each extension required additional scenario definitions, additional data and further experiments. For a graduation project with a fixed schedule, this supports defining a limited and clearly stated operating envelope from the start: Siyaq therefore restricts its initial scope recorded in clear weather and visibility conditions, and to two predefined scenarios, namely rear-end collisions and lane-change accidents.

The comparison with this line of work also clarifies that vehicle detection, movement analysis and traffic-event extraction are intermediate stages rather than final outputs. Fault cannot generally be inferred from the presence of vehicles in individual frames; the relevant evidence includes vehicle direction, changes in movement, the ordering of events, traffic-control information, and observed violations. This is reflected in the objectives defined for Siyaq, which cover vehicle detection and tracking, movement analysis, identification of relevant traffic events and predefined violations, and generation of supporting evidence from dashcam footage. The difference between the two systems is therefore one of context and integration rather than of underlying principle: Siyaq adapts the same evidence-to-rules pipeline to the two scenarios defined for this project, to Saudi traffic regulations, to one or two recordings of the same accident, and to a workflow in which an authorized expert issues the final decision.

Regulation-Grounded Fault Reasoning

Where the road rules in [10] and [11] are manually encoded for defined situations, more recent work has explored reasoning over regulatory knowledge together with accident evidence. Zhou and Zhang [12] introduced AITP (Artificial Intelligence Traffic Police), a multimodal large language model for traffic accident responsibility allocation. AITP combines multimodal reasoning with retrieval-augmented generation to incorporate regulatory knowledge and is accompanied by the DecaTARA benchmark for accident-reasoning tasks. As this work is currently available as a preprint, it is considered here as an indication of a recent research direction rather than an established result. Unlike AITP, which generates reasoning using retrieved regulatory knowledge, Siyaq uses a constrained set of explicit rules for predefined accident scenarios so that the reasoning can be inspected by an authorized expert. Al-Muzaini et al. [13] addressed a similar fault-assessment problem in the Saudi context through Muayen. Instead of accident video, Muayen uses textual descriptions provided by the two involved parties to predict bilateral fault-percentage splits on a 0%–100% scale built on the ratios of Regulation 2/60 [4]. The authors compared several classification approaches, with their hierarchical RoBERTa+ model achieving the best overall F1-score of 94.97%. Their labelling process was informed by Saudi traffic-assessment practice, the Saudi Traffic Law, and consultation with assessors familiar with Najm procedures. However, Muayen predicts fault from written accident narratives through a learned classification model rather than extracting evidence directly from accident footage and applying explicit rules at inference time. Siyaq instead uses video-derived evidence and explicit regulation-grounded rules so that the suggested percentage can be accompanied by its supporting evidence and reasoning for expert review. Taken together, the reviewed studies show a progression

from accident understanding toward explainable fault assessment. DRIVE [3] demonstrates temporal analysis and visual explanation from dashcam footage; Lee and Lee [7] show that accident sequences contain information relevant to negligence assessment; Wang et al. [8] demonstrate the extraction of scenario-specific manoeuvre evidence; and Yawovi et al. [10], [11] connect video-derived evidence to explicit road rules for responsibility assessment. AITP [12] extends this direction toward regulation-aware multimodal reasoning, while Muayen [13] demonstrates fault-percentage prediction in a Saudi assessment context using textual accident narratives.

Positioning of Siyaq and Research Gap

As the synthesis above shows, automated fault assessment has already been investigated, and Siyaq does not claim otherwise. However, within the literature reviewed for this project, the combination of automated dashcam-video analysis, explicit rule-based fault-percentage estimation grounded in Saudi traffic regulations, and mandatory review by an authorized expert is not directly addressed. The closest reviewed work in the Saudi context, Muayen [13], derives its fault percentages from the written accounts of the involved parties and identifies the analysis of accident images as future work, which leaves recorded footage as an evidence base for such an assessment largely unexplored. The contribution of Siyaq is therefore one of integration and adaptation rather than of a novel individual technique. Within a single constrained workflow, the system combines dashcam-based evidence extraction, vehicle detection and tracking, movement analysis, and the identification of predefined traffic events and violations. This evidence is then evaluated using explicit rules derived from relevant Saudi traffic regulations to produce a suggested fault percentage on a proportional scale of 25%, 50%, 75% and 100% (Regulation 2/60), with 0% for a party with no assigned fault [4], restricted to two supported scenarios: rear-end and lane-change accidents. One or two dashcam videos of the same accident are analyzed independently, and every suggested percentage is presented together with the detected events and the rules that were applied. The result remains subject to review, modification and approval by an authorized expert, positioning the system as decision support rather than a replacement for professional or legally authorized judgment. The reviewed literature also informs several design requirements. The manual inputs required by comparable responsibility-assessment systems [10] support minimizing the information users must provide, with selection of the applicable accident type identified as the primary manual input at this stage. The demonstrated feasibility of detecting lane-change manoeuvres from temporal driving video [8] supports treating manoeuvre and event identification as an intermediate output produced by the system rather than supplied by the user. The emphasis on interpretability across all reviewed approaches supports presenting every suggested fault percentage together with its supporting evidence and reasoning. The dataset limitations discussed above support treating data preparation as an explicit activity with defined selection criteria. Finally, the reported sensitivity of video-based systems to lighting and visibility conditions [10] suggests that an insufficient-evidence outcome should be supported when the uploaded footage does not provide adequate visual information, rather than forcing a potentially unreliable recommendation. These requirements will be refined during the requirements-definition and system-design stages.

3.2 Competitive Product Analysis

This section compares Siyaq with alternatives that address part of the same problem: the operational benchmark for accident-fault assessment in Saudi Arabia and global solutions for accident evidence analysis, collision reconstruction, liability estimation, and fault-ratio determination. Section 3.2.1 reviews the local solution, Najm, while Section 3.2.2 examines the identified global solutions. For each alternative, the discussion presents a description of the product, its main services, advantages, and disadvantages relative to the problem Siyaq addresses. Section 3.2.3 then compares the selected products with Siyaq using a feature-comparison table and discusses the main similarities, differences, and gaps identified through the comparison.

The selected products were not formally tested as part of this analysis. Most team members have previously used the Najm application; however, its accident-reporting workflow was not tested specifically for this study because it requires an actual accident case. Empiric Earth and Liablix are commercial platforms whose full accident-analysis workflows were not available to the team for direct testing. Although the Fault Ratio Information Portal is publicly accessible, it is based on Korean fault-ratio criteria rather than Saudi traffic regulations. Therefore, the comparison is based primarily on official product websites, published materials, and available documentation.

3.2.1 Local Solution

- **Najm**



Figure 5: Official Logo of Najm for Insurance Services. Source: Najm for Insurance Services [5].

The Najm application [5] is the official mobile application used for reporting and managing traffic accident cases in Saudi Arabia. Accident parties enter the required accident information through the application, while authorized assessors examine the accident information and available evidence and determine fault according to the Saudi Traffic Law and its Executive Regulations, which define proportional fault levels of 25%, 50%, 75% and 100% (Regulation 2/60). The application also offers remote services such as mobile damage assessment [5]. The Najm application represents the current local operational solution and is therefore the benchmark against which Siyaq is compared.

Main Services

- Reporting accidents through the Najm application by manually entering the required accident details, after which an authorized assessor is dispatched to the accident location.
- Displaying the fault percentages determined by the authorized assessor under the Saudi Traffic Law and its Executive Regulations.
- Providing access to the official accident report, which includes the accident details and assigned fault percentages.
- Providing remote damage assessment for eligible cases.

Advantages

- Operates directly within the Saudi regulatory context.
- Fault is determined by an authorized assessor.
- The issued outcome is official and legally effective.

Disadvantages

- Fault determination relies on manual human review rather than automated video analysis, and the accident details are entered manually by the accident party through the reporting form.
- An authorized assessor attends the accident location in many cases, although the Executive Regulations allow a simple accident between insured vehicles to be documented without attending the scene (Regulation 17/61) [4]; this can result in waiting time for the involved parties and obstruction of traffic flow. The Najm application therefore instructs drivers to move their vehicles to a safe location before completing the accident report when appropriate.
- The accident-reporting workflow does not provide an option for submitting dashcam footage for automated analysis as part of the fault-assessment process.
- The issued accident report presents the assigned fault percentage and a written description of the accident but does not provide a detailed explanation linking the available evidence and applicable traffic rules to the assigned fault percentage.

3.2.2 Global Solutions

- **Empiric Earth (Nexar and Nauto) — United States**



Figure 6: Empiric Earth Logo. Source: Empiric Earth [14]

Empiric Earth [14] is a United States-based commercial platform formed by combining Nexar and Nauto, two previously separate companies whose products are relevant to this project. Nexar's capabilities now contribute to World Intelligence, while Nauto's fleet-safety capabilities operate under Fleet Intelligence [14]. Nexar introduced a collision-reconstruction technology that uses raw camera data, on-board sensors, and deep learning algorithms to produce an automatically generated reconstruction immediately after a collision, presenting the accident on a map together with a textual timeline; the technology was deployed with the Japanese insurer Mitsui Sumitomo Insurance [15]. Nauto contributed incident-intelligence capabilities, where an on-device collision-detection model identifies real collisions, including low-impact incidents, and the platform assembles a first notice-of-loss report combining video, vehicle motion, driver behaviour, alerts, and sensor data with trip and environmental context [15]. The collision-reconstruction technology was announced by Nexar in 2021 [15]; the currently published material describes incident intelligence and first-notice-of-loss reporting under Fleet Intelligence [16].

Main Services

- Automatic collision reconstruction generated immediately after impact, with a map view and a textual event timeline [16].
- On-device collision detection, including low-impact incidents [15].
- Automated first-notice-of-loss reports assembled from road and in-cab video, vehicle motion and sensor data, driver behavior alerts, and environmental context such as road type, weather, road surface, and lighting [15].
- Retrieval of interior and exterior footage to investigate incidents and support driver exoneration [15].

Advantages

- Generates structured accident evidence directly from recorded footage and sensor data.
- Detects minor incidents that may otherwise go unreported.
- Proven commercial deployment with insurance providers and fleet operators.

Disadvantages

- No fault-percentage output and no explicit traffic-rule reasoning; the platform provides detected events, video, vehicle motion, and contextual information to support incident investigation and claims workflows rather than producing a proportional fault assessment.
- Fleet Intelligence relies on dedicated in-vehicle hardware, including an AI edge device with a dual-facing dash cam and on-device AI.
- Oriented toward commercial fleets and insurers rather than individual accident parties [15].
- Not adapted to the Saudi regulatory context.

- **Liablix — Italy**



Figure 7: Liablix Logo. Source: Liablix [16].

Liablix [17] is a commercial accident-reconstruction platform for insurers, appraisers, and legal professionals. It uses 3D simulation and physics-based modelling to reconstruct accident dynamics and provide information such as vehicle speeds, acceleration, trajectories, and other collision-related parameters (see Figure 8). According to information provided directly by Liablix, the platform does not currently estimate liability or fault percentages directly. Instead, it reconstructs accident dynamics based on physics and provides information that can subsequently be used to estimate liability. Liablix works with data submitted to the platform and can also take into account data originating from external hardware, such as telematics systems, black boxes, and dashcams, when such data are available. The platform provides several accident-analysis and reconstruction tools, as shown in Figure 9.

Main Services

- 3D reconstruction and simulation of accident dynamics (see Figure 8).
- Physics-based analysis of vehicle motion and collision behaviour.
- Analysis of information such as vehicle speeds, acceleration, and trajectories (see Figure 8).
- Processing of submitted accident data, including data from telematics systems, black boxes, and dashcams when available.
- Visualization of reconstructed collision dynamics and accident-analysis results (see Figures 8 and 9).

Advantages

- Provides detailed physics-based reconstruction of accident dynamics (see Figure 8).
- Provides quantitative information such as speeds, acceleration, and trajectories that can support subsequent fault assessment.
- Can incorporate data from different sources, including hardware-generated data when available.
- Provides multiple analysis and reconstruction tools within the platform (see Figure 9).

Disadvantages

- Does not currently determine fault percentages directly; instead, it reconstructs accident dynamics and provides information that can be used in a subsequent liability or fault assessment.
- Not adapted to the Saudi regulatory context and does not provide fault reasoning based on Saudi traffic regulations.



Figure 8: Liablix 3D Accident Reconstruction and Motion Analysis. Source: Liablix, reproduced with permission.

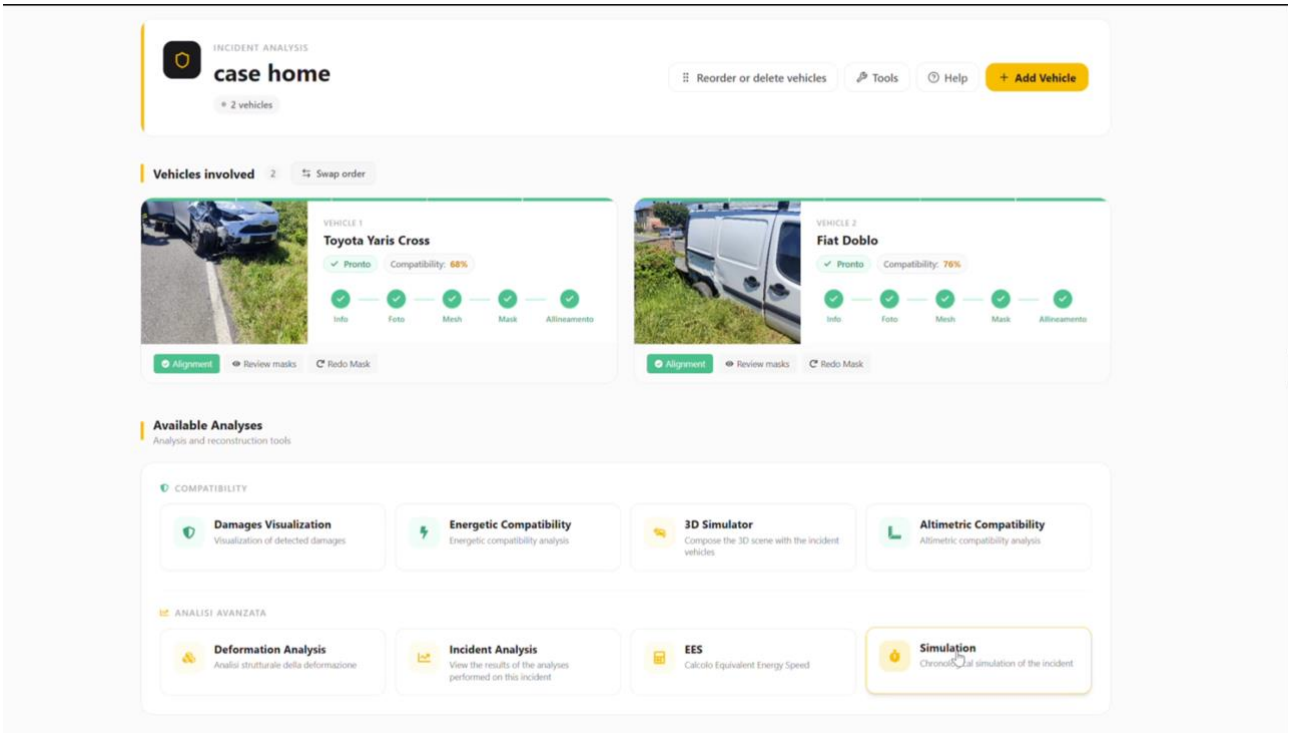


Figure 9: Liablix Incident Analysis and Reconstruction Tools. Source: Liablix, reproduced with permission.

- **Fault Ratio Information Portal - South Korea**



Figure 10: Fault Ratio Information Portal Logo. Source: General Insurance Association of Korea [17].

The Fault Ratio Information Portal [18] is a public service operated by the General Insurance Association of Korea that provides consumers with access to Korean automobile-accident fault-ratio criteria. Users can identify an accident scenario through the portal and view the corresponding basic fault ratio, together with information such as applicable laws, adjustment factors, and relevant precedents or mediation cases [18]. The portal provides predefined accident scenarios with basic fault ratios that can be adjusted according to specific circumstances; for example, individual scenarios may assign different basic ratios depending on vehicle movements and road conditions.

Main Services

- Provides predefined accident scenarios and their corresponding basic fault ratios.
- Presents adjustment factors that may increase or decrease the basic fault ratio depending on the circumstances of the accident.
- Provides applicable laws and relevant precedents or mediation cases alongside the selected accident scenario.
- Allows users to search for an accident scenario and review the corresponding fault-ratio information.

Advantages

- Provides fault ratios grounded in documented Korean fault-ratio criteria.
- Presents applicable legal information and adjustment factors alongside the selected accident scenario.
- Freely accessible to accident parties rather than restricted to insurers or professionals.

Disadvantages

- The accident scenario must be identified by the user; the portal does not analyse dashcam footage to automatically extract accident evidence or identify the applicable scenario.
- The returned value is based on predefined Korean fault-ratio criteria and the accident scenario selected by the user.
- Based on Korean fault-ratio criteria and therefore not directly applicable to the Saudi regulatory context.

3.2.3 Comparison and Discussion

The comparison focuses on features that reflect the main functions and objectives of Siyaq and distinguish it from the identified alternatives. The selected features cover the type of accident evidence used, the level of automation in evidence extraction and analysis, fault-percentage output, the basis and explainability of fault reasoning, the role of human review, and the operational scope of each solution. These features were selected because they represent the main aspects needed to compare how each alternative supports accident analysis and fault assessment and to identify the gap addressed by Siyaq. Table 5 summarizes the comparison of the selected alternatives based on these features.

Feature	Najm	Empiric Earth	Liablix	Fault Ratio Portal	Siyaq
Dashcam video as a primary analysis input	—	✓	—	—	✓
Automated evidence extraction from footage	—	✓	—	—	✓
Collision reconstruction or kinematic modelling	—	✓	✓	—	—
Proportional fault-percentage output	✓	—	—	✓	✓
Fault assessment grounded in documented national traffic rules/criteria	✓	—	—	✓	✓
Applied rule presented to the user alongside the evidence and result	—	—	—	Partial	✓
Authorized human review/decision required	✓	—	—	—	✓
Coverage of a wide range of accident types	✓	✓	✓	✓	—
Official, legally binding fault determination	✓	—	—	—	—

Table 5: Feature comparison of Siyaq and the identified alternatives.

The comparison shows that the reviewed alternatives address different parts of the accident-analysis and fault-assessment process. Najm provides official fault determination within the Saudi regulatory context but relies on manual review by an authorized assessor rather than automated video analysis. Empiric Earth focuses on automated accident evidence and incident intelligence, organizing information from video, vehicle motion, and other available data to support investigation and claims processes. Liablix focuses on physics-based accident reconstruction, providing information such as vehicle speeds, acceleration, and trajectories that can support subsequent fault assessment, but it does not directly determine a fault percentage. The Fault Ratio Information Portal provides predefined fault ratios based on documented Korean criteria and presents relevant legal information, but the accident scenario is identified by the user rather than automatically extracted from dashcam footage.

None of the reviewed alternatives combines automated extraction of accident evidence from dashcam footage with explicit fault reasoning grounded in Saudi traffic regulations. Siyaq is intended to integrate these two aspects by analyzing one or two dashcam recordings of the same accident

independently, automatically extracting relevant vehicle movements, traffic events, and violations, and evaluating this evidence using explicit rules derived from relevant Saudi traffic regulations. The system then produces a suggested proportional fault percentage together with its supporting evidence and reasoning for review by an authorized expert.

This integration is accompanied by deliberate scope limitations, as reflected in Table 5. Siyaq focuses on selected minor accident scenarios under clear weather and visibility conditions and does not perform physics-based collision reconstruction or kinematic modelling. Its purpose is therefore not to replace the authorized expert, but to provide an explainable, evidence-based recommendation that supports the expert's final review and decision.



4 Development of Siyaq AI Models

This chapter describes how the AI models of Siyaq are developed. Section 4.1 presents the dataset collected for the two supported accident types, its labels and how labeling is done. Section 4.2 presents the flow of the system components from the input video to the output, with the model intended for each component. Section 4.3 presents the rule-based system that turns the extracted evidence into a suggested fault percentage, based on the Saudi Traffic Law and its Executive Regulations [4].

4.1 Dataset

4.1.1 Dataset Source, Selection, and Preparation

Siyaq starts from the Car Crash Dataset (CCD) [22], [23], a public dataset built for traffic accident analysis. CCD contains real accident videos recorded by dashcams mounted on driving vehicles, which matches the input that Siyaq receives from its users. Its 1,500 accident videos were collected from YouTube channels and trimmed into short clips; each clip contains 50 frames at 10 frames per second, so every clip lasts five seconds. CCD also includes 3,000 normal-driving videos sampled from BDD100K, but Siyaq analyses accident videos only, so only the 1,500 accident videos are considered. The CCD annotation file gives, for each accident video, a binary label for each of its 50 frames (1 marks an accident frame), the start frame and the identifier of the source YouTube video, the timing (day or night), the weather (normal, snowy or rainy), and whether the ego-vehicle, meaning the vehicle that recorded the video, is involved in the accident [23].

CCD covers many accident types and conditions, while Siyaq supports only rear-end collisions and lane-change accidents (Section 1.3.4). CCD does not label the accident type, so the 1,500 videos could not be filtered by type automatically. The team therefore reviewed the clips manually and assigned each one to rear-end, lane-change, or neither. A clip is kept only if it shows a collision between two vehicles that fits one of the two accident types. Clips of other accident types, such as intersection, head-on or single-vehicle accidents, or accidents involving three or more vehicles, are removed . Clips in which the collision is not visible, or in which the events before the collision are missing, are also removed .

The filtering is still in progress, and the selected clips are kept together with their original CCD file name and YouTube identifier so that each one can be traced back to its source. The CCD repository is released under the MIT license, but the videos themselves were collected from YouTube; Siyaq therefore uses them only for academic development and evaluation.

The two supported accident types follow the definitions in Section 2.2. A rear-end collision is one in which the front of a vehicle hits the rear of another vehicle. A lane-change accident is a collision associated with a vehicle moving from one lane into another. These two types decide which events are labeled in each clip (Section 4.1.3) and which rules are applied later (Tables 9 and 10).

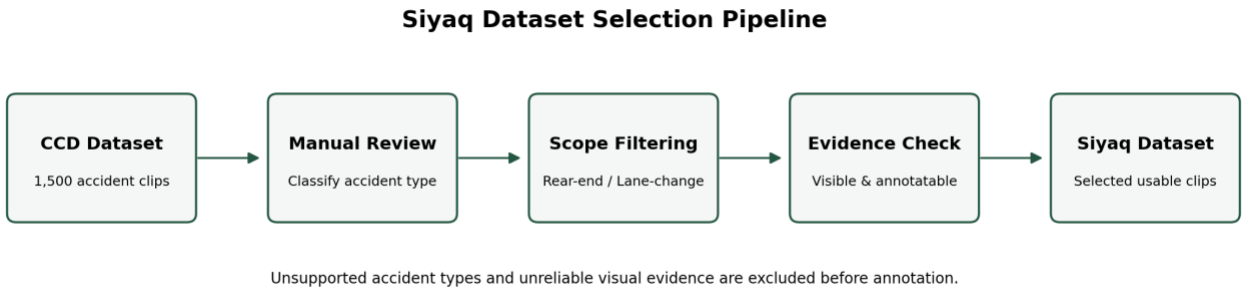


Figure 11: Dataset selection and filtering pipeline used for the Siyaq dataset.

4.1.2 Dataset Statistics

Table 6 shows the size of the Siyaq dataset at the time of writing. So far, 95 clips have been selected from the 1,500 CCD accident videos (about 6.3%): 57 lane-change clips (60.0%) and 38 rear-end clips (40.0%). All clips come from CCD and therefore have the same fixed length of 50 frames at 10 frames per second, so the average duration is 5.0 seconds for both accident types.

Accident type	Collected clips	Avg. duration (s)
Rear-end	*38*	*5.0*
Lane-change	*57*	*5.0*
Total	*95*	*5.0*

Table 6: Size of the Siyaq dataset by accident type.

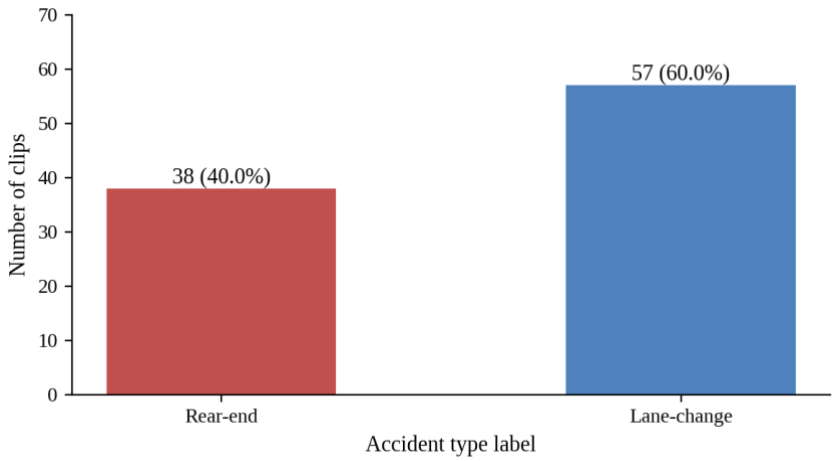


Figure 11: Label distribution in the Siyaq dataset.

4.1.3 Labels

The labels of the Siyaq dataset are grouped into three levels. Video-level labels describe the whole clip, object-level labels describe the vehicles and lane markings in each frame, and event-level labels

mark what happens over a range of frames. Each label is chosen because one of the Siyaq AI components, or the rule-based system, needs it as input or as ground truth for evaluation. Table 7 lists the labels, their meaning and the component that uses each one; the component names follow Table 8 and the rule codes follow Tables 9, 10 and 13. How the labels are created is described in Section 4.1.4.

Label	Meaning	Used by component
Accident type	Rear-end or lane-change, the scenario shown in the clip.	Event Analysis (decides which events to look for); Rule-Based Fault Estimation (selects the rear-end or lane-change rules)
~~Timing and weather~~	~~Day or night, and normal weather.~~	~~Rule-Based Fault Estimation (scope check)~~
~~Ego vehicle involved~~	~~Whether the vehicle that recorded the video is part of the collision. The recording vehicle is vehicle A in the rules.~~	~~Event Analysis; Rule-Based Fault Estimation~~
Vehicle	A bounding box around each vehicle in a frame, with an identity that stays the same across frames. The vehicles involved in the accident are marked as the recording vehicle (A) or the other vehicle (B).	Vehicle Detection & Tracking (YOLO + ByteTrack), for training and evaluation; Event Analysis
Lane marking	A polyline along each visible lane line, with its type: solid or dashed.	Lane Detection; Event Analysis (lane changes, L3 solid-line crossing)
Lane change	The start and end frames of a vehicle moving from one lane into another, with the vehicle and the direction (left or right).	Event Analysis (rules L1 to L6 and R3)
Turn signal	State of the turn signal of the lane-changing vehicle before the change: on, off or unclear.	Event Analysis (rule L2; also stated in the explanation of L1)
Sudden stop	The frames in which a vehicle brakes sharply, with the vehicle.	Event Analysis (rule R2)
Overtaking	The frames in which a vehicle passes another vehicle and returns to the lane.	Event Analysis (rule L5)
Collision	The frame of the first impact and the vehicles involved, with its visibility: clear, partly hidden or not visible.	Event Analysis; Rule-Based Fault Estimation (U5 when not visible)
Unclear / not visible	A value used instead of a label when an event cannot be judged reliably, such as a hidden collision, a signal that cannot be seen or lane markings that are not visible.	Rule-Based Fault Estimation (undetermined results)
Fault split	The expected fault split of the clip (vehicle A / vehicle B): one of 0/100, 25/75, 50/50, 75/25 or 100/0, or undetermined with its reason code.	Rule-Based Fault Estimation (ground truth for evaluation)

Label	Meaning	Used by component
Safe distant	Whether a safe following distance was kept before the collision, judged by whether the rear tires of the vehicle ahead are visible in the frame. Values: kept (tires visible), not kept (tires not visible) or unclear (cannot be judged).	Following Distance (tire detection), as ground truth for evaluation; Event Analysis (rules R1 and R2)

Table 7: Labels of the Siyaq dataset and the component that uses each label.

Label	Meaning	Values	Used by component
Accident type	The accident scenario represented in the video clip.	Rear-end / Lane-change	Rule-Based Fault Estimation
Vehicle	A bounding box around each vehicle in a frame, with an identity that remains consistent across frames. The vehicles involved in the accident are identified as vehicle A or vehicle B.	Vehicle A / Vehicle B / Other vehicle	Vehicle Detection and Tracking; Event Analysis
Lane marking	A polyline along each visible lane line and its type.	Solid / Dashed / Not visible	Lane Detection; Event Analysis
Lane change	Indicates whether a vehicle moves from one lane into another before the collision and the direction of the movement.	Left / Right / No lane change / Unclear	Event Analysis; Rule-Based Fault Estimation
Turn signal	Indicates the state of the turn signal of the lane-changing vehicle before the maneuver.	On / Off / Not visible	Event Analysis; Rule-Based Fault Estimation
Sudden stop	Indicates whether a vehicle brakes or stops suddenly before the collision.	Yes / No / Unclear	Event Analysis; Rule-Based Fault Estimation
Overtaking	Indicates whether a vehicle passes another vehicle before the collision.	Yes / No / Unclear	Event Analysis; Rule-Based Fault Estimation
Collision	Identifies the occurrence of the collision and whether the impact is visible in the video.	Clear / Partly hidden / Not visible	Event Analysis; Rule-Based Fault Estimation
Unclear / not visible	Indicates that an event or condition cannot be reliably determined from the available video evidence.	Unclear / Not visible	Rule-Based Fault Estimation
Fault split	The expected distribution of responsibility between vehicle A and vehicle B.	0/100 / 25/75 / 50/50 / 75/25 / 100/0 / Undetermined	Rule-Based Fault Estimation



Label	Meaning	Values	Used by component
Safe distance	Indicates whether a safe following distance was maintained between the vehicles before the collision.	Maintained / Not maintained / Unclear	Following-Distance Analysis; Rule-Based Fault Estimation

4.1.4 Labeling Process

The collected video clips were annotated using CVAT (Computer Vision Annotation Tool), which was selected to support the annotation of objects, lane markings, and traffic events in the dataset. Vehicle annotations were represented using bounding boxes, while lane markings were represented using polylines. Temporal events, such as lane changes, overtaking, sudden stops, and collisions, were annotated over the relevant frames. The annotations are exported in a format that can be processed and used with the selected AI models.

The collected clips were distributed among the four team members, with each member responsible for annotating her assigned videos. To improve the consistency of the annotations, the labeled clips are reviewed by a second team member after the initial annotation. Any disagreements or unclear annotations are discussed and resolved based on the agreed labeling definitions.

When an event cannot be determined reliably from the video, it is marked as unclear or not visible rather than assigning an unsupported label. For example, if a collision is partially hidden by another vehicle or object, the collision is not labeled as clearly visible unless sufficient visual evidence is available. Similarly, if a turn signal cannot be observed clearly, its state is treated as unclear. This approach helps reduce subjective annotations and prevents uncertain visual information from being treated as confirmed evidence.

4.1.5 Dataset Sample

(Raneem: the labeled frames from one video go here)



Figure 12: Annotated sample from the Siyaq dataset showing vehicle annotations and tracking across consecutive frames.

Figure 12 shows four annotated frames selected from one video clip in the Siyaq dataset. The frames demonstrate how vehicles are annotated using bounding boxes and tracked across different frames as their positions and sizes change. The annotations provide structured information about the vehicles present in the scene and their movement throughout the video.

System Components Flow

4.1.6 Flow Diagram

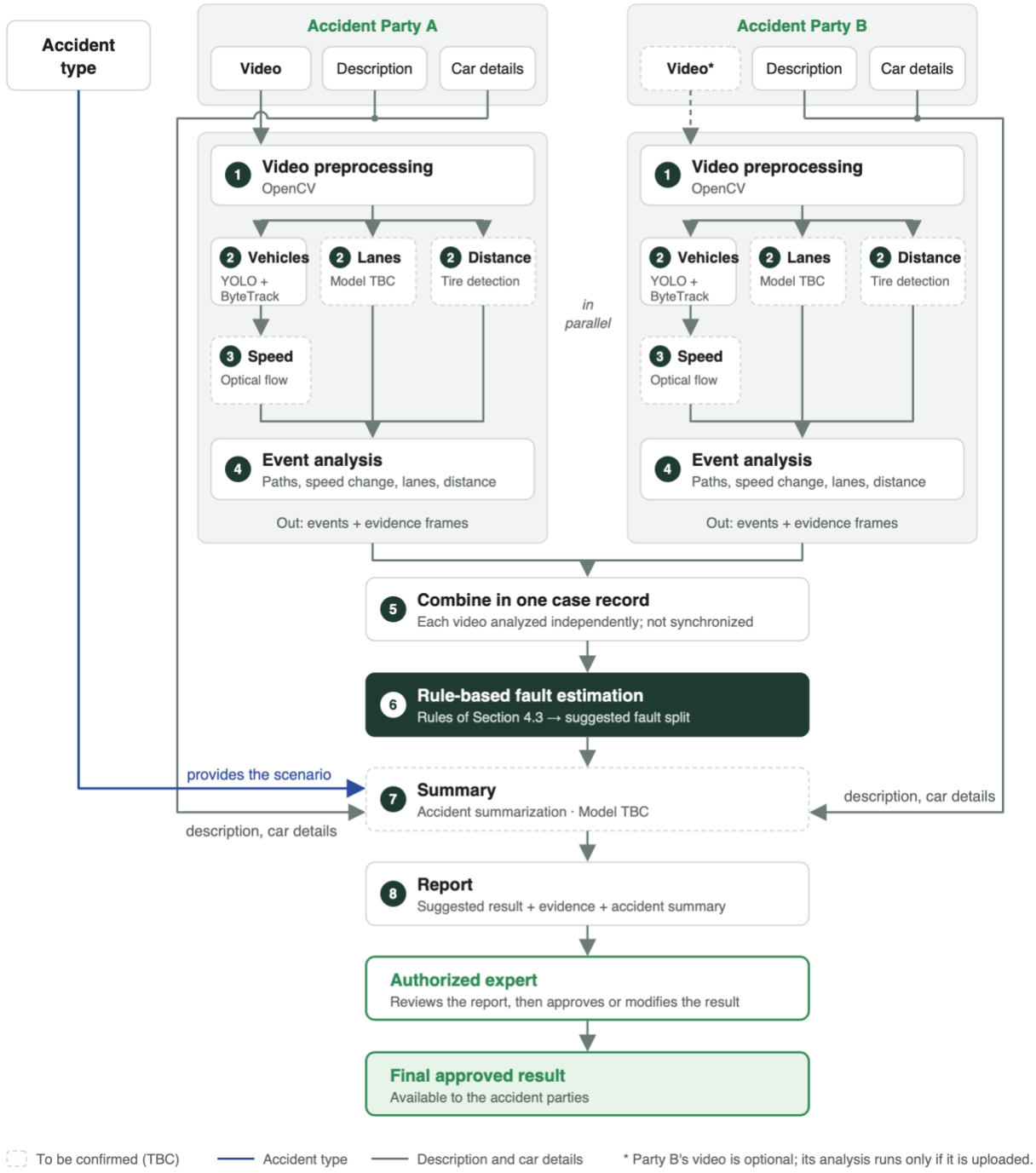


Figure 13: Flow of the Siyaq AI components from input to output.

4.1.7 Components Description

Following the flow shown in Figure 13, Table 8 describes each component of the Siyaq AI pipeline in the order the data moves through it, from the raw dashcam video to the final report presented to the authorized expert. For each component, the table states its intended model (where applicable), its input,

and its output. The draft report produced by the last component follows the information that an accident investigator records under Regulation 6/5/61: the location and time of the accident, the weather and road conditions, the number and condition of the parties, the vehicles involved and the location of their damage, and how and why the accident occurred [4].

Component	Intended model	Input	Output
Video Preprocessing	OpenCV	Raw dashcam video (Party A / Party B)	Processed video frames
Vehicle Detection & Tracking	YOLO + ByteTrack	Processed frames	Detected vehicles per frame, tracked across frames (position/gap between vehicles)
Lane Detection	[TBC]	Processed frames	Lane-line position and type (solid/dashed)
Distance Detection	Tire detection	Processed frames	Distance between vehicles
Speed Detection	Optical flow	Processed frames	Vehicle speed and speed change (acceleration/deceleration) over time
Event Analysis	[TBC]	Vehicle paths, speed change, lane-line data, and distance data	Detected traffic events such as lane change, sudden stop, and following distance between vehicles + evidence frames
Combine in One Case Record	Logic step (no ML model)	Events/evidence from Party A's and Party B's streams (each analyzed independently, not synchronized)	One case record combining both parties' evidence
Summary	[TBC]	Combined case record + accident type + descriptions + car details (Party A and Party B)	Accident summary
Rule-Based Fault Estimation	Rule-based (Section 4.3)	Accident summary	Suggested fault split (or undetermined, Section 4.3.3)
Report	[TBC]	Suggested result + evidence + accident summary	Final report (sent to Authorized Expert)

Table 8: Siyaq AI components with their intended models, inputs and outputs.

4.2 Rule-Based System

The rule-based system receives the evidence extracted by the AI components for one case and returns either a suggested fault percentage for each party or an undetermined result. The rules are derived from the Saudi Traffic Law and its Executive Regulations [4]. All provision numbers in this section refer to that document and are read from right to left: the last number is the article, and each number to its left is an item within the one to its right. For example, 2/1/1/6/50 is item 2 of Regulation 1/1/6/50.

"Schedule" refers to the numbered violation and fine schedules attached to the Law; for example, Schedule 2, item 11 is item 11 of Violations Schedule No. (2).

4.2.1 Fault Rules and Examples

The Executive Regulations define four fault ratios: 100%, 75%, 50% and 25% (Regulation 2/60). Responsibility arises when there is a direct or indirect causal link between the act and the accident (Regulation 1/60), and the ratio is set according to three grounds: negligence, lack of caution, and failure to comply with the regulations (Regulations 1/2/60 to 3/2/60) [4]. As in Section 2.1.4, Siyaq also uses 0% for a party with no assigned fault. The Regulations do not assign a ratio to any specific accident scenario, and they do not state how the ratios are divided between two parties. The ratios in Table 9 and Table 10 are therefore not taken from the Regulations; they are a design decision of Siyaq, set by the following mapping principle:

- If only one party committed a violation that caused the collision, that party is assigned 100% and the other party 0%.
- If both parties committed violations and one violation is judged to be the main cause, the ratio is 75% / 25%.
- If both parties committed violations that contributed equally, the ratio is 50% / 50%.

The mapping principle is applied once, when each rule is written, so that every rule has one fixed ratio; the system does not apply it at run time. In each rule, the legal part is the provision that the party violated, while the ratio is Siyaq's suggestion and remains subject to review by the authorized expert (Section 2.5.2). Table 9 lists the rules for rear-end collisions and Table 10 lists the rules for lane-change accidents.

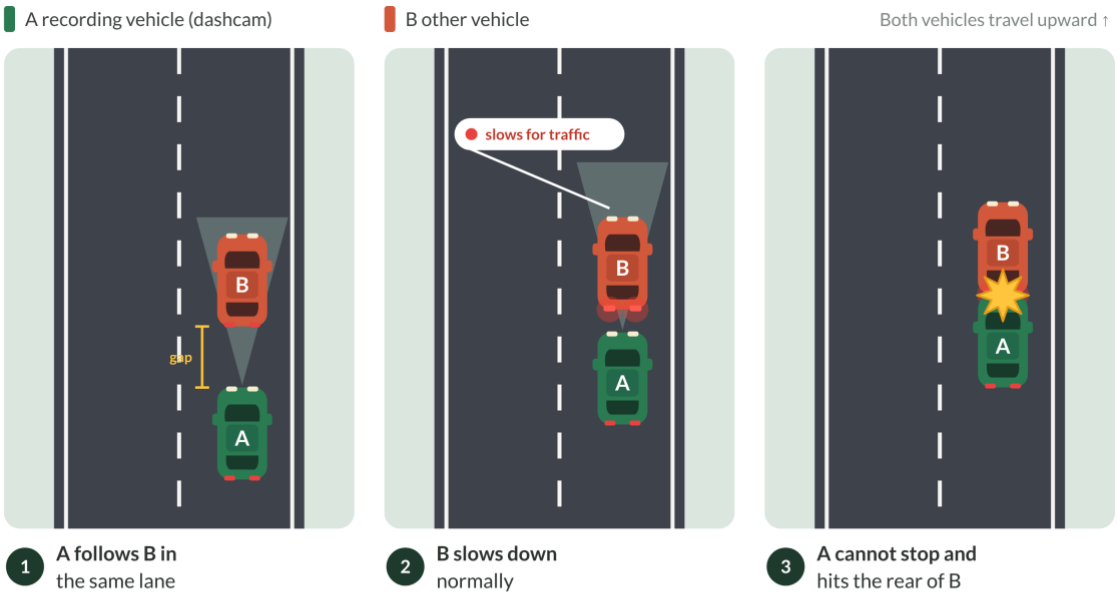
Rule ID	Situation	Evidence required	Provision violated [4]	Suggested ratio (rear / front)
R1	Rear vehicle hits a front vehicle moving or stopped normally in the same lane	Both vehicles in the same lane before impact; no lane change or sudden stop by the front vehicle; gap closing until impact	Rear: 19/2/50 (distance that increases with speed), 1/9/50 (speed that allows stopping in an emergency); 3/9/50 (reduce speed in congested traffic); Schedule 2, item 11. The front vehicle, as the leading vehicle, has priority (1/1/1/6/50)	100% / 0%
R2	Front vehicle stops suddenly with no traffic reason while the rear vehicle is too close	Sharp deceleration of the front vehicle; no visible reason ahead (obstacle, queue or signal); short gap before the stop. Sudden braking is allowed only for necessity or safety (5/2/50)	Front: 5/2/50 and 1/1/1/6/50 (no sudden stop); Schedule 3, item 5. Rear: 19/2/50 and 1/2/50 (traffic rules do not relieve the driver of the duty of care)	75% / 25%
R3	Front vehicle entered the rear vehicle's lane shortly before impact	Front vehicle crosses the lane line within a short time before impact	Routed to the lane-change rules; the entering vehicle must not change direction suddenly (1/1/1/6/50)	See L1–L3

Table 9: Fault rules for rear-end collisions. Ratios follow Siyaq's mapping principle, not ratios stated in [4].

Rule ID	Evidence variable	Required value / condition
R1	**Same-lane movement**	Both vehicles are detected in the same lane before the collision
	Front vehicle motion	The front vehicle is stopped or moving slower than the rear vehicle before impact
	Pre-collision maneuver	No sudden lane entry or sudden braking is observed immediately before impact, where assessable.
	Following distance	Gap is below the defined safe-distance threshold before impact
	Collision	Detected between the rear and front vehicles
R2	**Same-lane movement**	Both vehicles are detected in the same lane before the collision
	Following distance	Gap between the vehicles is below the defined safe-distance threshold before impact
	Front vehicle deceleration	A sudden decrease in the front vehicle's motion is detected before the collision
	Collision	Detected between the rear and front vehicles
R3	**Lane entry**	Front vehicle crosses the lane boundary and enters the rear vehicle's lane before impact
	Time interval	Time between lane entry and collision is below a defined threshold
	Following distance	The gap immediately after lane entry is below the defined safe-distance threshold
	Collision	Detected between the involved vehicles

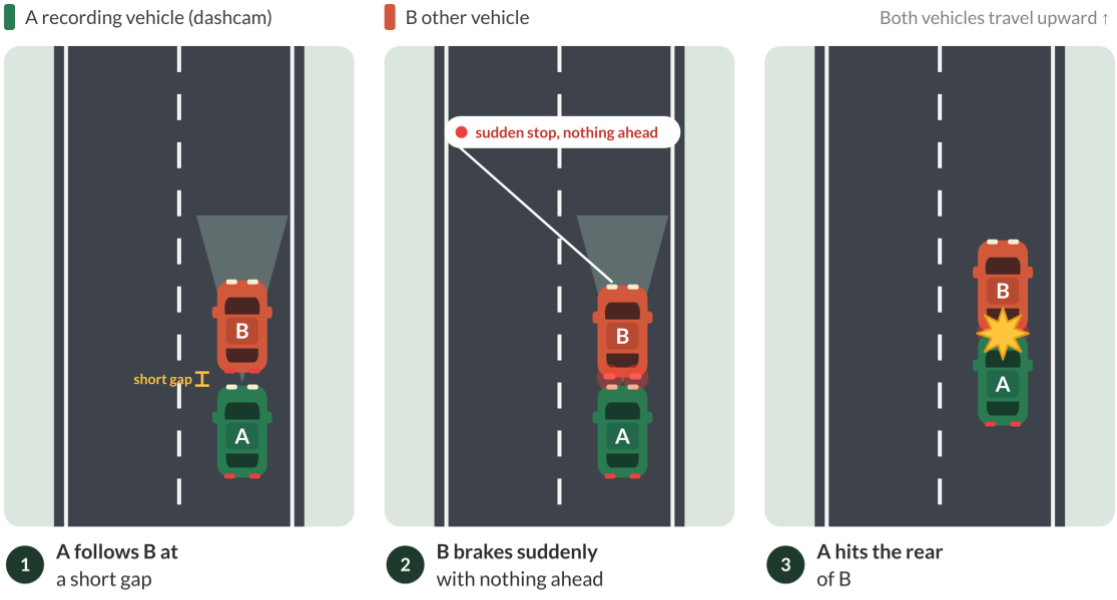
Table 10: Rear-End Collision Rules - Evidence Variables and Required Conditions

Figure 14 and Figure 15 illustrate rules R1 and R2 using hypothetical examples. In all rule figures, vehicle A (green) is the vehicle that recorded the video and vehicle B (red) is the other vehicle.



AI EVIDENCE	RULE APPLIED	SUGGESTED RATIO	FINAL DECISION
Same lane, no lane change B slows gradually Gap closes until impact No sudden stop by B	**R1** A did not keep a safe distance (19/2/50) or a speed that allows stopping (1/9/50) [4]	**A 100%** **B 0%** Siyaq mapping, not stated in [4]	Authorized expert reviews, may modify, and approves

Figure 14: Hypothetical example of rule R1: the rear vehicle hits the front vehicle, which slows down normally in the same lane.



AI EVIDENCE	RULE APPLIED	SUGGESTED RATIO	FINAL DECISION
Sharp deceleration of B No obstacle, queue or signal Short gap before the stop Same lane, no lane change	**R2** B stopped suddenly (5/2/50, 1/1/1/6/50); A too close (19/2/50) [4]	**A 75%** **B 25%** rear / front · Siyaq mapping	Authorized expert reviews, may modify, and approves

Figure 15: Hypothetical example of rule R2: the front vehicle stops suddenly with no traffic reason while the rear vehicle follows too closely.

Rule ID	Situation	Evidence required	Provision violated [4]	Suggested ratio (changing / other)
L1	Vehicle changes lane into a vehicle already going straight in that lane	Lane-line crossing; other vehicle alongside or behind in the target lane, going straight	Changing: 2/1/1/6/50 (priority to the vehicle going straight), 13/2/50 and 1/24/2/50 (make sure the change is safe); Schedule 4, item 15	100% / 0%
L2	Same as L1, with no turn signal	As L1; turn signal off	Also 1/7/50 and 2/24/2/50 (signal in good time); Schedule 2, item 1	100% / 0%
L3	Same as L1, across a continuous (solid) lane line	As L1; the crossed line is continuous	Also 9/7/6/2/6/50 (no driving on or crossing continuous longitudinal lines, listed under the overtaking rules); Schedule 3, item 7 (not keeping within the marked lanes)	100% / 0%
L4	Vehicle entering or leaving a free-flow road from or to a service road collides with a vehicle going straight	Entry from, or exit to, a service road; other vehicle going straight	3/1/1/6/50 (priority to the vehicle going straight); no dedicated schedule item	100% / 0%
L5	Overtaken vehicle speeds up while the overtaking vehicle returns to the lane	L5a: the overtaking vehicle had completed the pass (its rear ahead of the other vehicle's front) when it started to return, and the overtaken vehicle accelerated. L5b: the overtaking vehicle returned before completing the pass	Overtaken: 4/2/6/50 (must not speed up). Overtaking: 5/6/2/6/50 (sufficient lateral distance); no dedicated schedule item	L5a: 25% / 75%; L5b: 75% / 25%
L6	Two vehicles move into the same lane at the same time	Both vehicles crossing toward one lane	Both: 13/2/50	50% / 50%

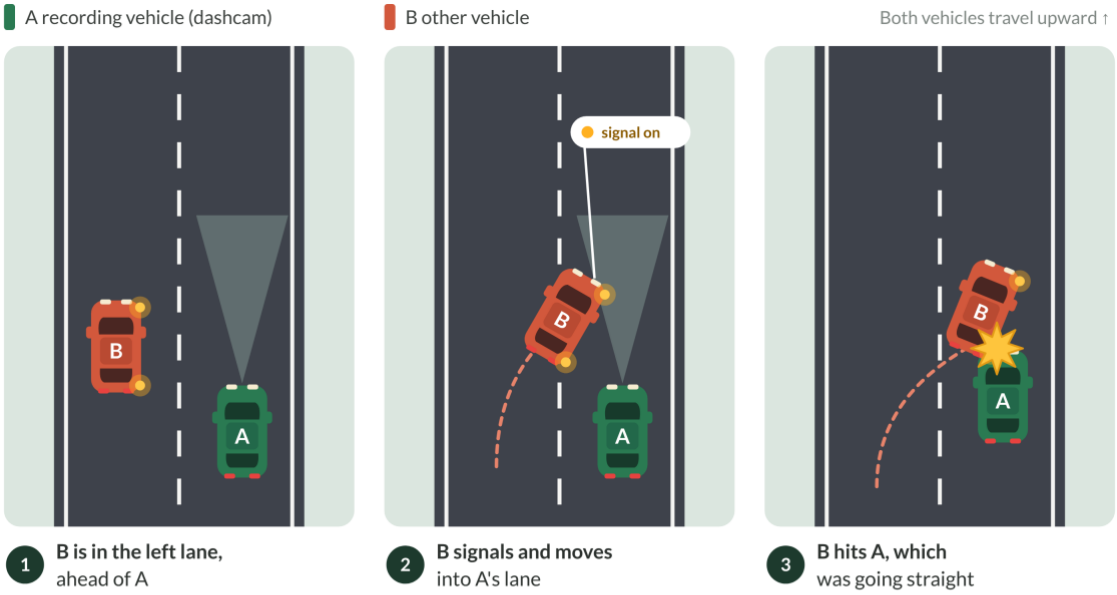
Table 11: Fault rules for lane-change accidents. Ratios follow Siyaq's mapping principle, not ratios stated in [4].

Rule ID	Evidence variable	Required value / condition
L1-L2-L4	**Lane change**	A vehicle crosses the lane boundary and enters an adjacent lane
	Unsafe gap	The available longitudinal or lateral gap between the vehicles is below the defined safety threshold during the lane change
	Collision	Detected between the involved vehicles during or shortly after the lane change
	Lane-line type	Dashed
L3	**Lane change**	Detected

	Lane-line type	Solid
	Collision	Detected between the involved vehicles
L4	Vehicle Deceleration	The entering vehicle reduces its speed before entering the main road.
	~~Safe Entry Gap~~	~~A sufficient gap is available in the main road traffic to allow the entering vehicle to merge without creating a collision risk.~~
	~~Collision~~	~~A collision occurs during entry onto the main road or immediately after the entry maneuver.~~
L5a	**Overtaking completed**	The overtaking vehicle has moved fully ahead of the other vehicle before returning to its original lane.
	Return to original lane	The overtaking vehicle starts moving back toward the lane it was driving in before overtaking.
	Other vehicle acceleration	The overtaken vehicle increases its speed while the overtaking vehicle is returning to its lane.
	Event sequence	The overtaking is completed first, followed by the return maneuver and acceleration of the overtaken vehicle.
	Collision	Detected between the overtaking vehicle and the overtaken vehicle during the return to the original lane.
L5b	**Overtaking Incomplete**	The overtaking vehicle has not moved fully ahead of the other vehicle.
	Early Return	The overtaking vehicle starts returning to its original lane while the other vehicle is still alongside it or ahead of it.
	Event Sequence	The return maneuver begins before overtaking is completed.
	Collision	Detected between the overtaking vehicle and the overtaken vehicle during the incomplete overtaking maneuver or early return to the original lane.
L6	**Simultaneous Lane Change**	Two vehicles move toward the same lane at approximately the same time.
	Collision	The two vehicles collide during or immediately after the lane changes.

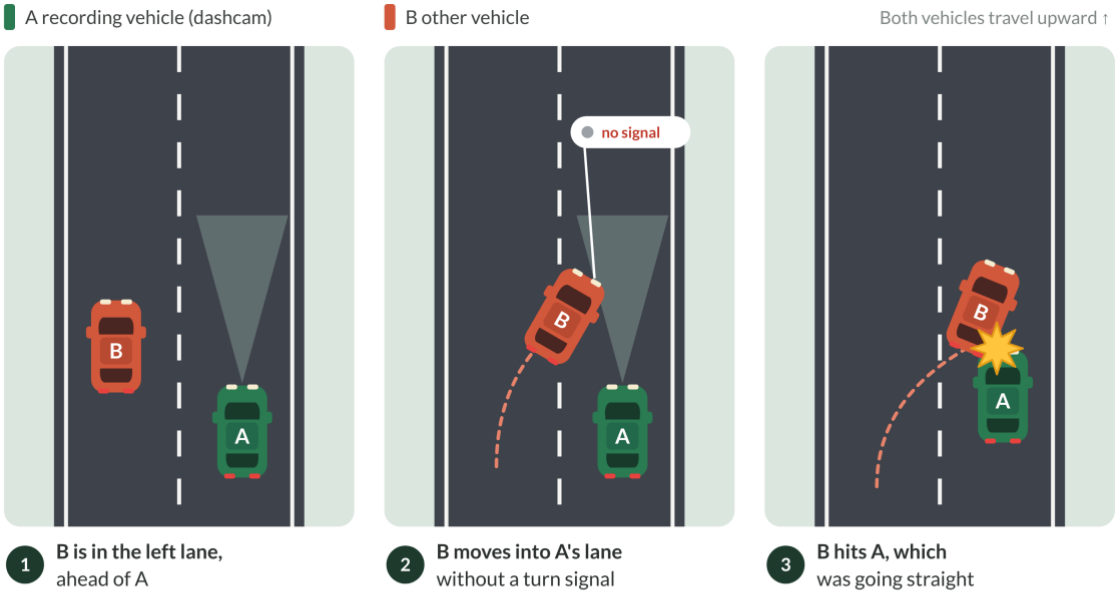
Table 12: Lane Change Collision Rules - Evidence Variables and Required Conditions

Figures 16 to 21 illustrate rules L1 to L6 in the same way.



AI EVIDENCE	RULE APPLIED	SUGGESTED RATIO	FINAL DECISION
Lane change by B A going straight in its lane B's signal on B alongside, cuts in	**L1** B did not give priority to the vehicle going straight (2/1/1/6/50, 13/2/50) [4]	**B 100%** **A 0%** Siyaq mapping, not stated in [4]	Authorized expert reviews, may modify, and approves

Figure 16: Hypothetical example of rule L1: a vehicle changes lane, with its signal on, into a vehicle going straight.



AI EVIDENCE	RULE APPLIED	SUGGESTED RATIO	FINAL DECISION
Lane change by B A going straight in its lane B's signal off B alongside, cuts in	**L2** As L1, and B did not signal in good time (1/7/50, 2/24/2/50) [4]	**B 100%** **A 0%** Siyaq mapping, not stated in [4]	Authorized expert reviews, may modify, and approves

Figure 17: Hypothetical example of rule L2: as L1, but the changing vehicle does not use its turn signal.

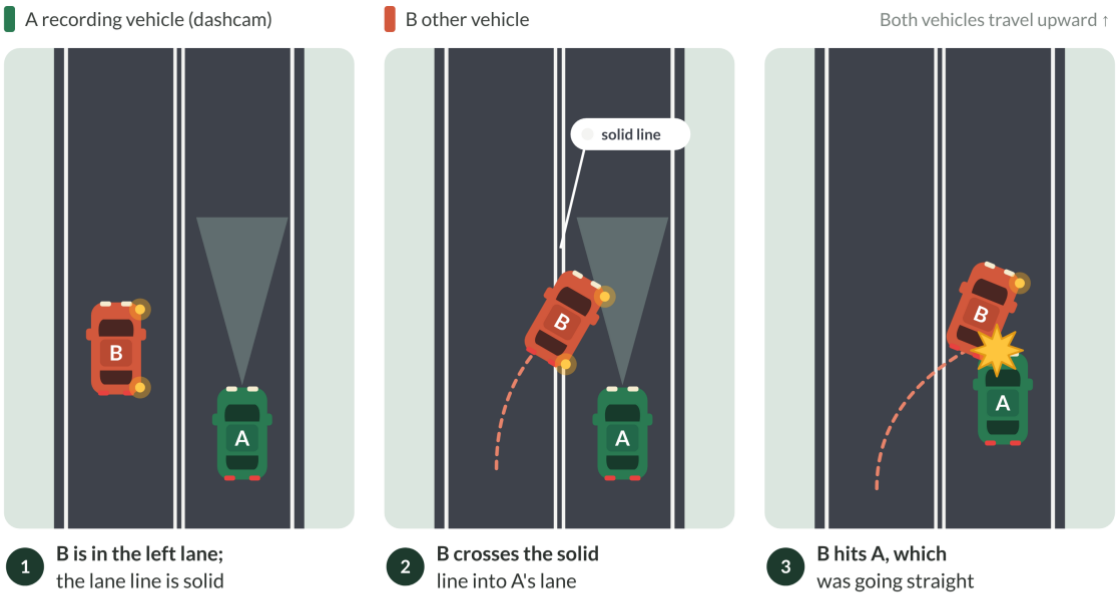


Figure 18: Hypothetical example of rule L3: as L1, but the changing vehicle crosses a solid lane line.

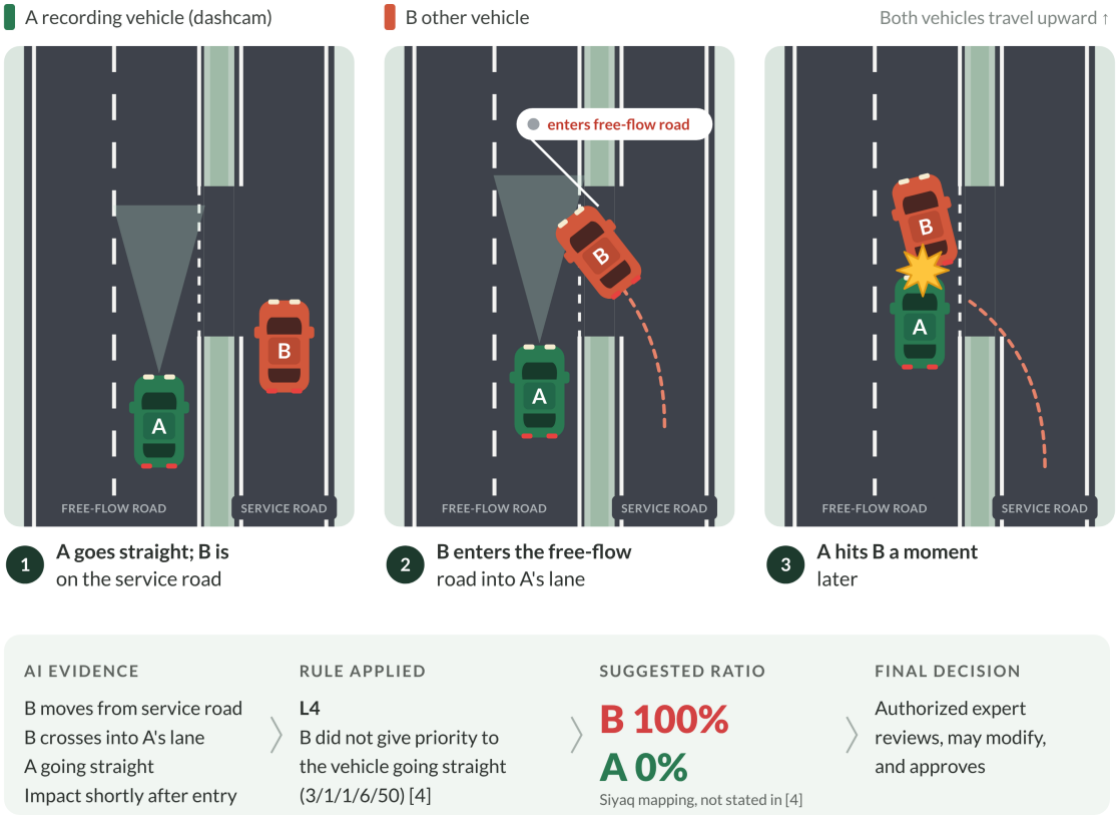


Figure 19: Hypothetical example of rule L4: a vehicle enters the free-flow road from the service road into a vehicle going straight.

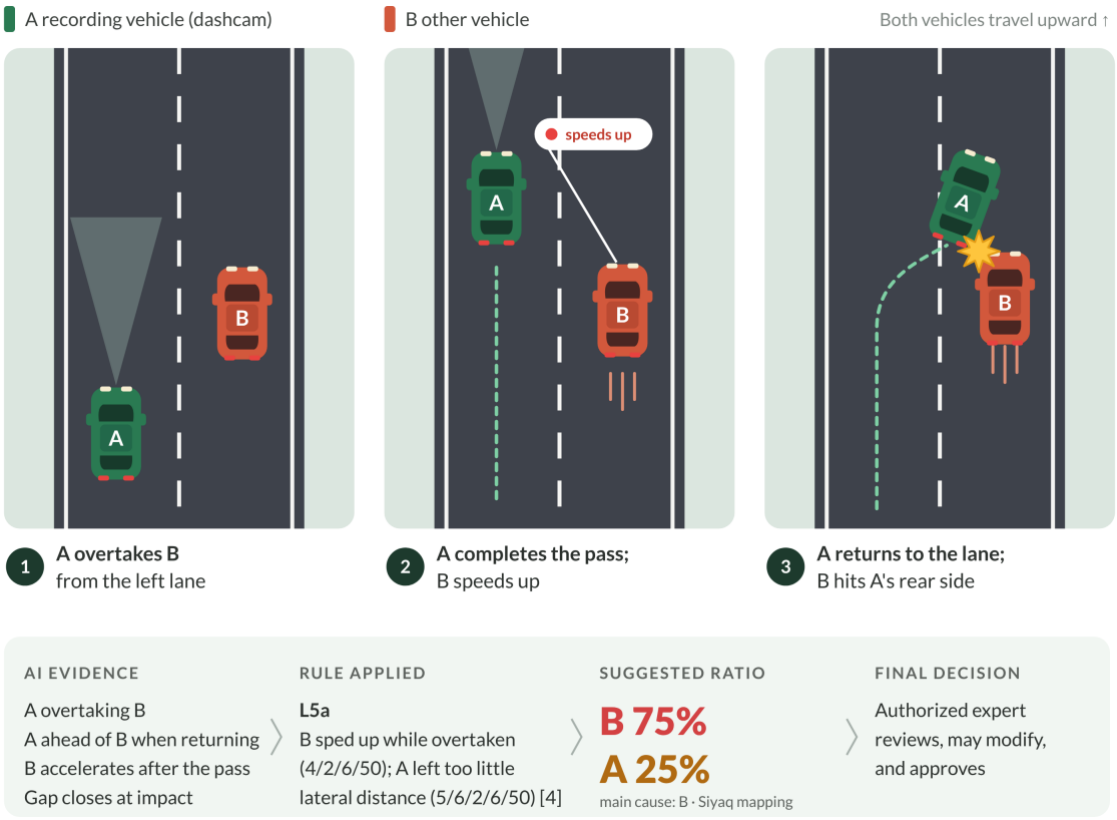


Figure 20: Hypothetical example of rule L5a: the overtaking vehicle has completed the pass when the overtaken vehicle speeds up.

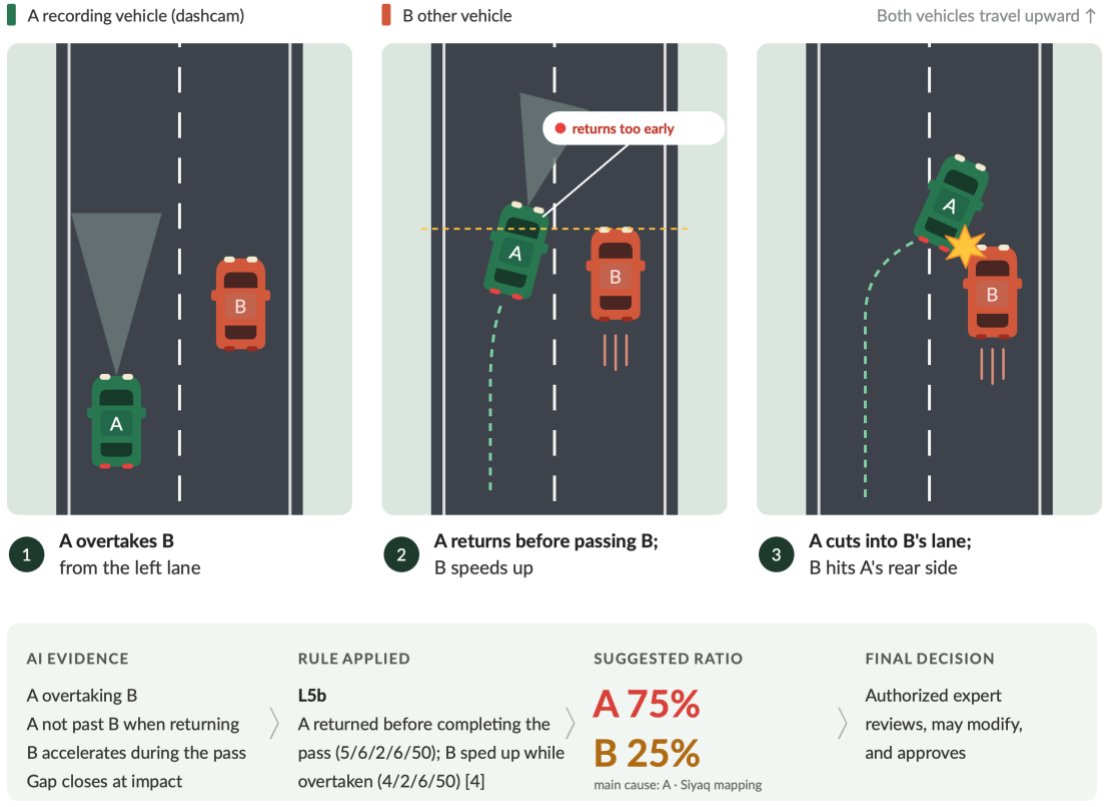


Figure 21: Hypothetical example of rule L5b: the overtaking vehicle returns before completing the pass while the overtaken vehicle speeds up.

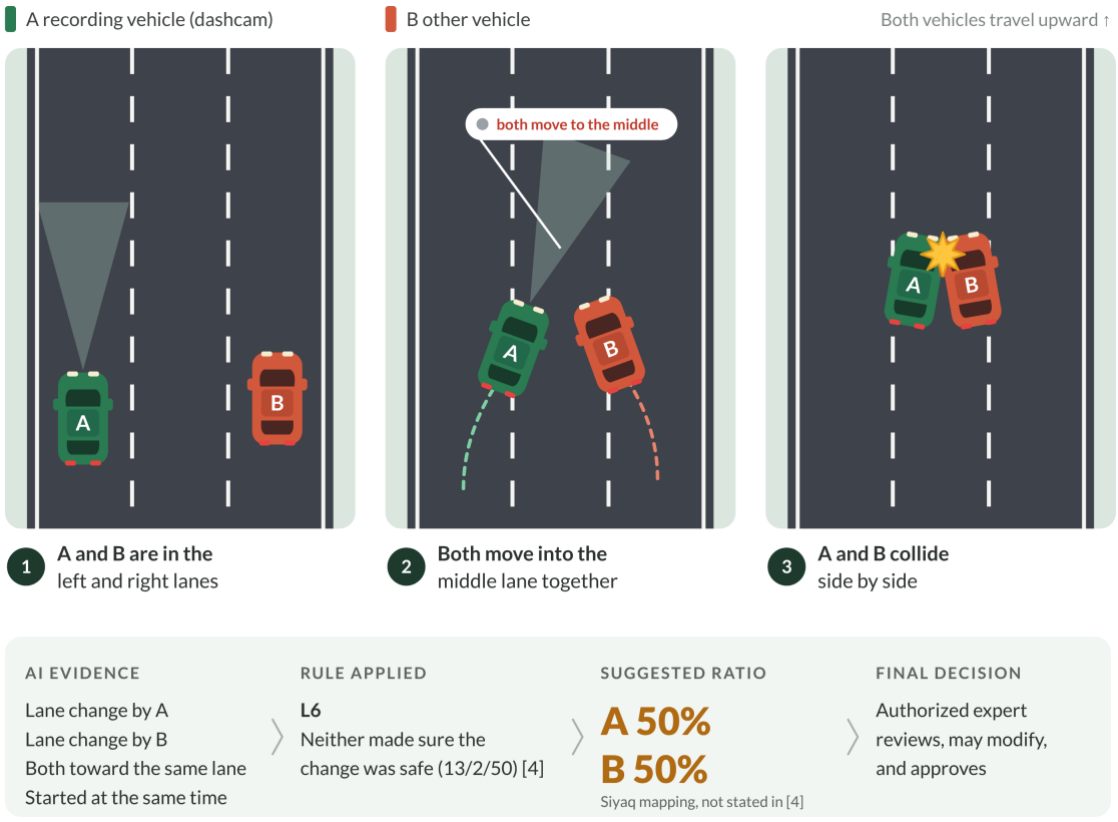


Figure 22: Hypothetical example of rule L6: two vehicles move into the same lane at the same time.

Rules L2 and L3 add a violation to L1 but do not change the ratio, because in L1 the changing vehicle is already the only party that violated a provision. The ratios of R2, L5 and L6 follow the mapping principle. In L5 both vehicles violate a provision, so the main cause is decided by the order of events: if the overtaking vehicle had completed the pass and the overtaken vehicle then accelerated (L5a), the overtaken vehicle is the main cause; if the overtaking vehicle returned before completing the pass (L5b), the overtaking vehicle is the main cause. If the overtaken vehicle did not accelerate, the case is L1, and if the order of events cannot be established, the result is undetermined (U10). Some evidence in Tables 9 and 10 depends on capabilities of the AI components that are defined in Section 4.2: the turn-signal state (L2), the type of lane line (L3), whether there is a reason ahead for a sudden stop (R2), and the thresholds for a "short gap" and "a short time before impact" (R1 to R3). A rule whose evidence cannot be extracted cannot be applied automatically. The turn signal must be given in good time before the move (Regulation 2/24/2/50), so if the signal state is detected, it is checked over a time window before the lane line is crossed; if it is not detected, the case is still decided under L1 (Section 4.3.3). The side of a lane change is not a violation by itself, because overtaking on the right is allowed when the road has more than two lanes in one direction (Regulation 5/2/6/50) [4].

Example (hypothetical). A dashcam clip shows vehicle B in the right lane, moving alongside the recording vehicle and slightly ahead of it. B crosses the lane line into the recording vehicle's lane with no visible turn signal, and the recording vehicle hits the rear of B shortly after. The AI components report a lane change by B, the turn signal off, and the recording vehicle going straight in its lane. Because B entered the lane shortly before impact, rule R3 routes the case to the lane-change rules, and L2 matches: B did not give priority to the vehicle going straight (2/1/1/6/50) and did not signal

(1/7/50). The suggested result is 100% on B and 0% on the recording vehicle, even though the impact looks like a rear-end collision. The authorized expert reviews this suggestion before the final decision. Figure 22 illustrates the example.

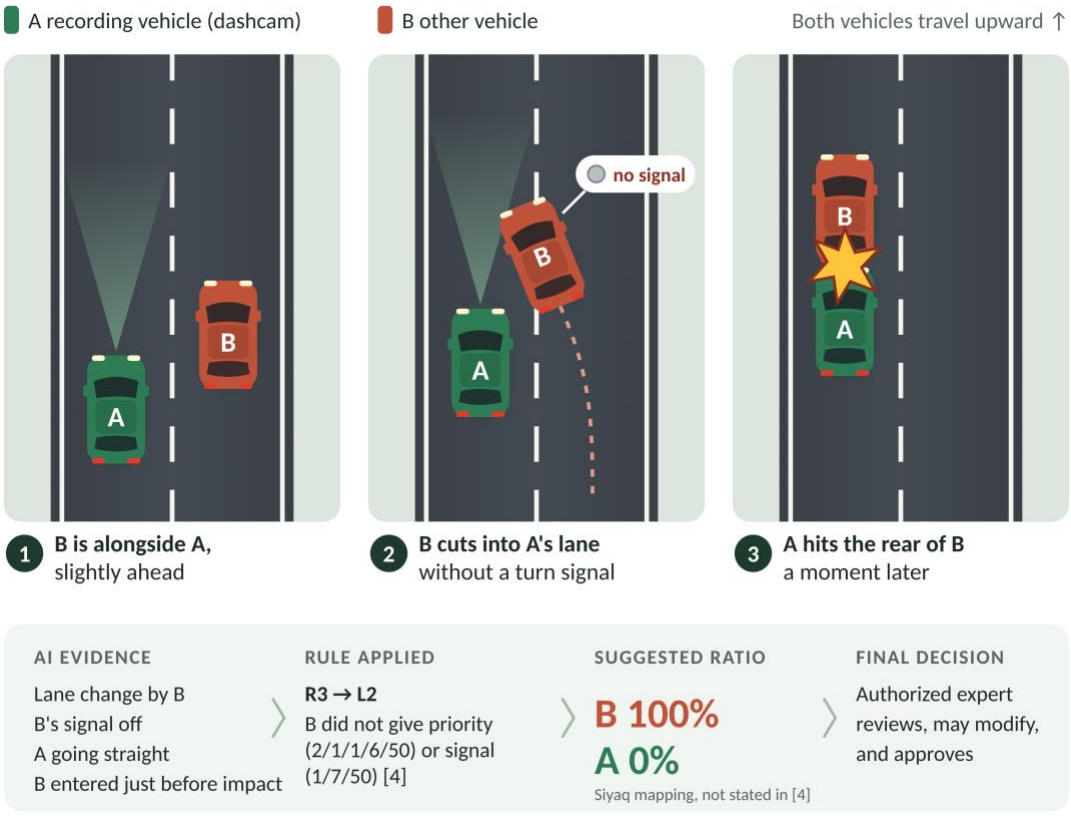


Figure 23: Hypothetical example: rule R3 routes a rear impact after a late lane change to rule L2.

4.2.2 Incorporating the Traffic Rules

Siyaq incorporates the Saudi traffic rules as a fixed set of rules written in advance from the Traffic Law and its Executive Regulations [4]. The rules are not learned from data: the AI components only extract evidence from the video, and the rule-based system interprets that evidence (Section 2.5). This section explains how the provisions were selected, how each provision becomes a rule, and how the rules are applied to a case.

The provisions were selected by reading the Law and its Executive Regulations and keeping only those that apply to the two supported scenarios, to responsibility and fault ratios, and to the situations in which no ratio can be suggested. Table 11 lists the groups of provisions used.

Group	Provisions [4]	Used for
Responsibility and fault ratios	1/60; 2/60 (1/2/60 to 3/2/60); 3/60	Causal link, the four ratios and the grounds of responsibility
Accident classes	1/59, 2/59, 3/59 (3/3/59)	Scope (U3) and intentional accidents (U12)
Following distance and speed	19/2/50, 1/9/50, 3/9/50, 1/2/50	R1, R2

Group	Provisions [4]	Used for
Braking and the leading vehicle	5/2/50, 1/1/1/6/50	R1, R2, R3
Lane change and signalling	13/2/50, 1/24/2/50, 2/24/2/50, 1/7/50, 2/1/1/6/50, 3/1/1/6/50, 5/2/6/50	L1, L2, L4, L6
Overtaking and lane lines	4/2/6/50, 5/6/2/6/50, 9/7/6/2/6/50	L3, L5
Special cases	4/60, 5/60, 6/60, Article 63(2)(b)	U13 to U15 and the meaning of an undetermined result
Violation and fine schedules	Schedule 2, items 1 and 11; Schedule 3, items 5 and 7; Schedule 4, item 15	The violation recorded for each rule

Table 13: Provisions of the Traffic Law and its Executive Regulations used by Siyaq.

Regulation 3/60 describes responsibility by the act that was committed or omitted and caused the accident, classified by the grounds in Regulation 2/60 [4]. Each Siyaq rule follows this: it records the act (its conditions), the provision violated and the ground. Because every rule is tied to a written provision, the ground recorded is failure to comply with the regulations (3/2/60); negligence and lack of caution (1/2/60 and 2/2/60) are not assessed separately, since they depend on judgement that the video evidence does not show. Table 12 shows the structure of a rule, using rule L2.

Field	Content	Example (rule L2)
Rule ID	Unique code	L2
Accident type	Rear-end or lane-change	Lane-change
Conditions	Evidence that must be present, extracted by the AI components (Section 4.2)	Vehicle B crosses the lane line; turn signal off; other vehicle going straight in the target lane
Provision	Article, regulation and schedule item violated [4]	2/1/1/6/50, 1/7/50, 2/24/2/50; Schedule 2, item 1; Schedule 4, item 15
Ground	One of the grounds in 2/60	Failure to comply with the regulations (3/2/60)
Suggested ratio	Set by the mapping principle (Section 4.3.1)	B 100%, other vehicle 0%
Explanation	Text shown to the expert with the evidence frames	Vehicle B changed lane without signalling and did not give priority to the vehicle going straight (2/1/1/6/50, 1/7/50).

Table 14: Structure of a Siyaq rule, with rule L2 as an example.

The rules are applied to a case in the following order, as shown in Figure 23:

1. **Scope check.** The case is checked against U1 to U3 (Table 13). If any applies, the result is undetermined.
2. **Evidence extraction.** Each submitted video, the mandatory video of the first party and the optional video of the second party, is analyzed independently by the AI components (Section 4.2), which output the events observed in that video.
3. **Evidence check.** If the events of a video show one of U4 to U8, that video cannot support a suggestion.

4. **Combining the videos.** The events from the available videos are placed together in one case record; Siyaq does not fuse the videos (Section 3.1). If the two videos give inconsistent evidence, the result is undetermined (U9), and the discrepancy is shown to the expert instead of being resolved by an automatic rule.
5. **Rule matching.** Rule R3 is checked first: if the front vehicle entered the lane a short time before impact, the lane-change rules are used. The conditions of each rule are then compared with the events. If exactly one rule matches and it has a ratio, it gives the suggestion; if no rule matches, or rules with different ratios match, the result is undetermined (U10, U11).
6. **Result for the expert.** The result, whether a suggested ratio with its rule, provision and explanation, or an undetermined result with its reason code, is stored with the case and presented to the authorized expert, who reviews it, may modify the ratio, and approves the final result (Section 2.5.2).

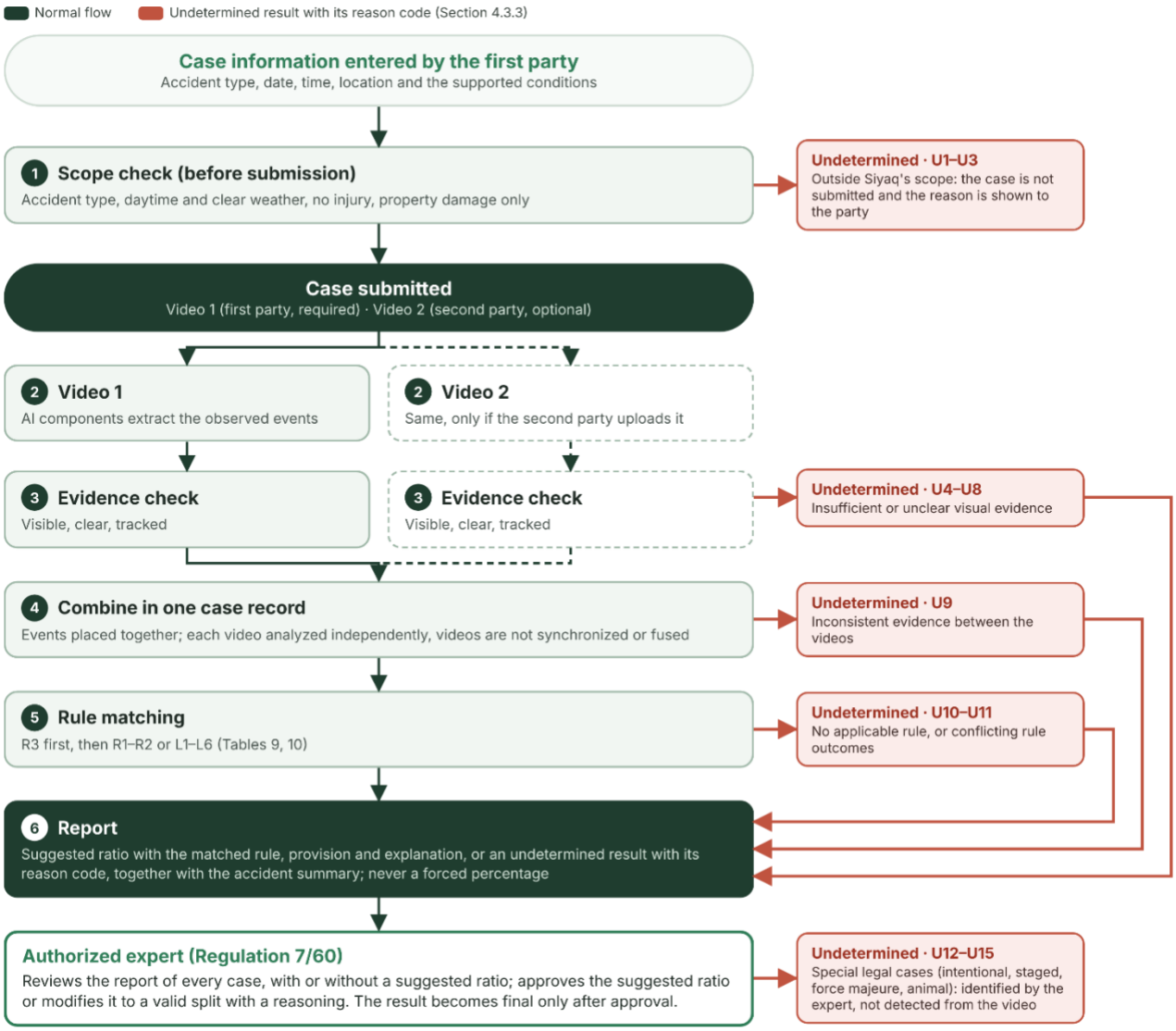


Figure 24: Application of the Siyaq rules to a case, with the points where an undetermined result is returned.

This keeps the final determination with a human, as in the Regulations, where the fault ratio is determined by the accident investigator under the supervision of the competent officer (Regulation 7/60) [4]. Siyaq's suggestion supports this determination and does not replace it.

The rules and their ratios are planned to be evaluated against assessments by authorized accident assessors; the team intends to approach Najm for Insurance Services for this purpose. In the planned evaluation, the assessors would assess a set of accident clips that were not used in building the rules, covering both supported scenarios, without seeing Siyaq's result, and their fault ratios would then be compared with Siyaq's suggestions. The comparison would check, separately, whether the evidence was extracted correctly, whether the suggested ratio matches the assessor's ratio, and whether the undetermined results were justified. If the evaluation is carried out, its results will be reported in a later sprint.

4.2.3 Situations When the Fault Percentage Cannot Be Determined

When any situation in Table 13 applies, Siyaq does not suggest a fault percentage. It returns an undetermined result with the reason code and the available evidence, and the authorized expert completes the assessment without a suggested percentage. An undetermined result is not a legal finding: under Regulation 6/60, it is the competent authority that, after completing all inference and inspection procedures, refers a case to the competent court when responsibility cannot be determined [4]. Siyaq's undetermined result only means that the system cannot support a suggestion with its rules and evidence. The situations are grouped into four categories: outside Siyaq's scope, insufficient or unreliable visual evidence, no deterministic rule, and special legal cases.

Code	Category	Situation	Why no percentage can be given	Basis
U1	Outside scope	Accident type other than rear-end or lane-change (e.g., intersection, head-on, single-vehicle, three or more vehicles)	No rules exist for these types	Section 1.3.4
U2	Outside scope	Recorded at night or in adverse weather (e.g., rain, fog, sandstorm)	Outside the scope and the dataset conditions	Section 1.3.4
U3	Outside scope	Any injury, or major damage to property	Siyaq covers no-injury, property-damage-only accidents; this excludes minor accidents with injury (2/59) and serious accidents (3/59)	Section 1.3.4; 2/59, 3/59, 3/3/59 [4]
U4	Visual evidence	Events before the collision are not in the clip	The lane change, signal and gap cannot be judged, so the causal link cannot be shown	Section 2.3.3; 1/60 [4]
U5	Visual evidence	The collision is not visible (out of frame or blocked)	The point of impact and the vehicles involved are unknown	Section 2.3.3; 1/60 [4]
U6	Visual evidence	Low resolution, heavy blur, dropped frames or strong glare	The frames cannot be trusted as evidence	Siyaq design
U7	Visual evidence	A vehicle is lost by tracking or detected with low confidence	The vehicle's path before impact is incomplete	Siyaq design

Code	Category	Situation	Why no percentage can be given	Basis
U8	Visual evidence	Lane markings are not visible	A lane change cannot be told apart from movement within a lane (R1, R3, L1 to L6). The Law defines a lane even when it is not marked (Article 2), so this is a technical limitation of Siyaq	Section 2.3.3
U9	Visual evidence	The two submitted videos give inconsistent evidence	The events of the two videos cannot be placed in one consistent account	Section 3.1
U10	No deterministic rule	No rule matches the evidence, or the matching rule has no agreed ratio (e.g., L5 when the order of events cannot be established)	No rule gives a suggestion	Siyaq design
U11	No deterministic rule	Two or more rules match with different ratios	The rule base has no single answer	Siyaq design
U12	Special legal case	The collision appears intentional	An intentional accident is not a traffic accident under 1/59 and is investigated by the police	1/59 [4]
U13	Special legal case	The collision appears staged	Staging a traffic accident is an offence	Article 63(2)(b) [4]
U14	Special legal case	The accident is caused by force majeure beyond the driver's will	The driver is exempted from the public right	4/60 [4]
U15	Special legal case	An animal is a party to the accident	Its owner or keeper bears responsibility if this is proven	5/60 [4]

Table 15: Situations in which Siyaq does not suggest a fault percentage.

U1 is known from the accident type selected by the accident party. How U2 and U3 are checked, whether by a statement in the case form or by an automatic check, is not established yet and will be defined with the flow in Section 4.2. U4 to U9 are detected from the outputs of the AI components, and U10 and U11 by the rule-based system. U12 to U15 are not detected from the video; they are listed because, when they apply, the rules of Section 4.3.1 do not apply and the case is left to the authorized expert.

A turn signal that cannot be seen does not by itself prevent a suggestion. Rule L1 does not depend on the signal, so when the other conditions of L1 are met, the suggestion is made under L1 and the explanation states that the signal state was not observed. When the party selected a rear-end collision but rule R3 routes the case to the lane-change rules, the explanation shown to the expert states this change.

5 System Description

5.1 Users

Accident Party

The primary users of the proposed system are individuals involved in minor traffic accidents who want to submit their accident cases for analysis. These users may have different educational backgrounds and varying levels of technical expertise. They are not required to have prior knowledge of traffic accident assessment or artificial intelligence. Instead, they are expected to have basic computer skills and be familiar with using web applications. Through the system, accident parties can submit the required accident information, provide available dashcam footage, and review the final approved report.

Authorized Expert User

The authorized expert users are qualified professionals responsible for reviewing traffic accident cases submitted through the system. They are expected to have relevant educational qualifications and practical experience in traffic accident assessment, as well as a good understanding of the Saudi Traffic Law and its Executive Regulations. Expert users review the generated accident report and its supporting evidence, including the accident frames identified by the system. They can approve the generated report, modify the fault percentage or accident description when necessary. In terms of technical expertise, expert users are only expected to have basic computer skills and be familiar with using web-based systems. The system provides a simple and clear interface that allows experts to review cases and supporting evidence without requiring advanced technical knowledge.

5.2 Requirements Elicitation

The requirements of Siyaq were elicited using multiple sources and methods to obtain data. A questionnaire was conducted with 40 participants to collect information about users' experiences, needs, expectations, and opinions regarding traffic accident assessment and the proposed system. The participants had different age groups, driving experience, and levels of experience with web applications and dashcam recordings, and included both individuals who had previously experienced traffic accidents and those who had not.

In addition, the team conducted informal discussions and inquiries with a small number of experts in traffic accident investigation and assessment. These experts provided domain-specific and practical information based on their experience in investigating and assessing traffic accidents. The discussions were used to understand the current assessment process, the information and evidence considered during accident investigation, and the requirements that should be considered when designing the proposed system.

The team also reviewed relevant documents and files provided through these expert discussions. These documents were used as an additional source of information to understand the procedures, information, and rules related to traffic accident assessment. The collected information from the questionnaire,

expert discussions, and document analysis was then used to identify and define the functional and non-functional requirements of Siyaq.

5.2.1 Questionnaire Results

The questionnaire was completed by 40 participants with different demographic and experience backgrounds. Most participants were either 18–24 years old or 45 years old and above, and the participants included both males and females. Most participants had several years of driving experience and had an advanced level of experience with web applications. The participants also had different levels of experience with downloading or transferring dashcam recordings, allowing the team to consider different user capabilities when defining the system requirements.

The questionnaire showed that 34 out of 40 participants (85%) had previously experienced a traffic accident. Among those participants, most accidents were assessed through the attendance of a specialist at the accident location, while some were assessed remotely. The assessment process commonly took one to three hours, while other cases took less than one hour or longer. This indicated a need for a faster and more efficient accident assessment process. Therefore, Siyaq should support an electronic and streamlined assessment process that reduces unnecessary waiting and processing time.

The responses also identified several problems with the current accident assessment process. Participants who were dissatisfied mainly mentioned long waiting times, difficulty providing evidence, lack of clarity regarding the reason for the assigned fault percentage, and delays in receiving the final result. These findings led to several requirements for Siyaq. The system should provide an easy method for uploading and submitting evidence, clearly explain the basis of the generated result, and provide the user with the result without unnecessary delays.

Dashcam recordings were also an important area of the questionnaire. Although only 14 participants (35%) currently use a dashcam, the majority of participants recognized its value in accident assessment. 32 participants (80%) stated that dashcam recordings greatly help clarify the circumstances of traffic accidents, while the remaining participants stated that they help to some extent. This finding supports making dashcam video a central input for Siyaq. The system should therefore allow users to upload dashcam recordings and use the available video evidence to identify relevant vehicles, movements, and accident events.

The questionnaire also showed a positive level of acceptance toward the proposed system. 34 participants (85%) either supported or strongly supported the idea of Siyaq, while 27 participants (67.5%) stated that they would prefer to use it if they experienced a simple traffic accident. This indicates a user need for an accessible electronic solution that allows accident parties to submit their cases without relying entirely on in-person procedures. Accordingly, Siyaq should provide a simple web-based process for submitting accident information and dashcam evidence and receiving the final result electronically.

The participants' responses regarding the most important system features provided direct input for defining the functional requirements. Fast accident analysis and displaying the evidence used in the analysis were the most frequently selected features, with 30 participants (75%) selecting each. This supports the requirement for Siyaq to process the accident within a defined target time and to present the final report along with evidence supporting its analysis.

An easy-to-use interface was selected by 24 participants (60%), indicating that usability should be considered when designing the system. Therefore, the system should provide a clear and simple Arabic interface and minimize unnecessary steps when submitting and reviewing a case. Easy video uploading was selected by 23 participants (57.5%), supporting the requirement for a straightforward video-upload process. In addition, 16 participants (40%) selected the ability to track the case status, supporting the inclusion of a feature that allows users to follow the progress of their case.

5.2.2 Informal Expert Consultation

The team conducted informal discussions and inquiries with a small number of experts in traffic accident investigation and assessment to gain practical and domain-specific knowledge about the accident assessment process. The information collected through these discussions helped the team understand the current workflow, the evidence used during accident investigation, the process of determining fault percentages, and the information included in the final accident report.

One of the main findings was that there is a focus on reducing the accident analysis time to approximately 20 minutes. Siyaq sets a stricter target of 10 minutes from case submission until the expert receives the generated report (PBI 30). The experts also indicated that rear-end collisions and lane-change accidents represent a large proportion of traffic accidents in Saudi Arabia, estimated at around 80%. Based on this information, the team decided to focus the initial system scope on these two accident types rather than attempting to cover a wide range of accident scenarios. This allows the system to focus its analysis and rule-based logic on a limited number of clearly defined scenarios.

The discussions also provided information about the current direction toward remote accident assessment. The assessment process is increasingly being conducted remotely, with the final report expected to be sent within less than 24 hours. This supported the decision to develop Siyaq as a web-based system**,** allowing the accident party to submit the required information and dashcam evidence electronically without requiring an in-person process.

The experts also explained the general structure of the final accident report provided to the user. The report includes information about the involved parties, accident details, insurance information, the assigned fault percentage, and a description of the accident. This information helped the team identify the main content that should be considered when designing Siyaq's final report.

Privacy was also identified as an important consideration. The other party's identifying information in the final report is currently masked to protect their privacy, while their insurance information is not included in the masking. This information helped the team define privacy-related requirements for the information displayed to users and consider how information belonging to the other party should be presented.

Regarding the expert's role, the discussions indicated that the generated result should be presented to the expert in the form of a ready report to reduce the time required for assessment. The expert can then approve or modify the fault percentage and the accident description. The evidence and frames illustrating the accident are provided as attachments to support the report rather than being part of the main report content. These findings directly influenced the design of the expert interface and the review workflow in Siyaq.

The experts also explained that when dashcam footage is available during an accident investigation, the parties' statements may also be collected; however, dashcam footage is given greater importance because it provides visual evidence of the accident. This supports the requirement for Siyaq to use the dashcam video as a major source for identifying the sequence of events, vehicle movements, and relevant evidence.

Finally, the discussions provided important information about how fault percentages are determined. The assessment relies heavily on information and rules contained in traffic-system files and related regulations [4]. Therefore, these sources will be used as a major basis for developing Siyaq's rule-based responsibility estimation component, rather than relying only on the AI model to determine the result.

Overall, the expert consultation helped the team identify requirements related to supported accident types, analysis time, remote assessment, report contents, privacy, expert review, evidence presentation, and responsibility estimation. These findings were used together with the questionnaire results and the reviewed documents to define the system requirements and scope.

5.3 Architecture

Siyaq follows a client-server architecture in which the presentation layer is separated from the backend processing and from the external data-storage services. This separation assigns distinct responsibilities to the client, backend, analysis, and storage components, while allowing the Server to coordinate their interactions. Figure 25 shows the major subsystems of the system, the external device, external services, and user roles that interact with it, and the way requests and data flow between them over the Internet.

The system has two user roles: an accident party (User) and an authorized Expert, both of whom interact with the system through the Siyaq Web App. Since an accident typically involves two parties, the User who submits a case can provide the email address of the second accident party, and the system sends them an invitation link through the External Messaging API so they can join the same case. The first accident party is required to upload a single dashcam video when submitting the case, while the second accident party may optionally upload their own video when completing their portion of the case, so a case includes one or two dashcam videos depending on whether the second party provides one. A dashcam is the external device considered by the system. It is not directly connected to Siyaq; instead, it is used to record accident footage that is later uploaded by the User through the web application as part of a case submission.

The Siyaq Web App is the client-side component and the main point of interaction for both user roles. Through the application, the User creates an accident case, submits the required information, and uploads a single recorded dashcam video. The Expert uses the same application to review the generated analysis, modify the suggested fault percentage if necessary, and provide final approval. Requests from the web application are sent to the backend Server over the Internet, and the corresponding responses are returned through the same communication path. The only exception is the dashcam video itself, which the web application uploads directly to the External Video Storage after receiving a signed upload authorization from the Server.

The Server is the central backend component of the system and internally organizes its responsibilities into two modules: an Application Logic module and an AI Processing & Fault Analysis module. It receives the requests sent from the web application, coordinates the overall case workflow, stores and retrieves case data through the External Database, manages the uploaded videos in the External Video Storage, and passes the video(s) internally to its AI Processing & Fault Analysis module for analysis.

The External Messaging API is an external service that Siyaq uses to send email messages to users. The Server requests it to deliver a one-time verification code (OTP) during account registration, a password-reset link when a user requests to reset their password, and an invitation link when the User provides the second party's email address. The second party opens the invitation link through the Siyaq Web App to join the same accident case and optionally upload their own dashcam video.

The External Map Service (OpenStreetMap and Nominatim) is used by the Siyaq Web App to let the User select the accident location on a map. The Web App displays the map tiles from OpenStreetMap, and when the User selects a point, it requests the location name from Nominatim. The selected coordinates and location name are then sent to the Server and stored with the case.

The External Database (Neon) is a cloud-hosted PostgreSQL service and the shared data repository of Siyaq. It is accessed only by the Application Logic module and stores the information required throughout the accident-case lifecycle, including users, accident cases, video identifiers, analysis results, and expert decisions. The video files themselves are not stored in the database.

The External Video Storage (Cloudinary) is a cloud service that stores the uploaded dashcam videos. After a video is uploaded, the Application Logic module verifies the upload with Cloudinary and records the video's identifier in the External Database. When a video needs to be viewed, the Server generates a short-lived signed link so that only authorized users can access it, and the video is then delivered from Cloudinary directly to the web application. The Server also retrieves videos from Cloudinary when they are needed for analysis.

The AI Processing & Fault Analysis module, maintained within the Server, performs the main video-based analysis required to assess an accident case. At a high level, it performs vehicle detection, vehicle tracking, and movement analysis, and matches the extracted information against the predefined rules derived from the Saudi Traffic Law and its Executive Regulations [4]. Once the second accident party completes and submits their required information, the case becomes ready for analysis. The Application Logic module then retrieves the first party's mandatory video and, if provided, the second party's optional video from the External Video Storage and passes them to this module for analysis. The module then returns an analysis result containing the supporting evidence and a suggested fault percentage to the Application Logic module, which stores it in the External

Database and presents it through the web application for expert review.

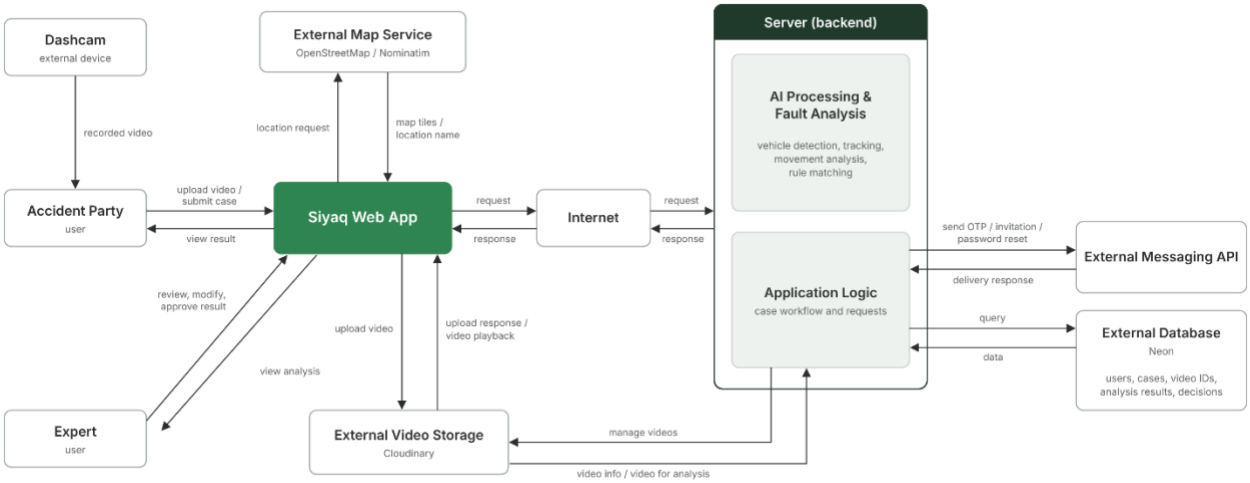


Figure 25: System Architecture Diagram of SIYAQ Web App.

5.4 Use Case Diagram

The Use Case diagram below *(See Figure 25)* provides an overview of the Siyaq system, highlighting its main functionalities and the interactions between the system and its different users.

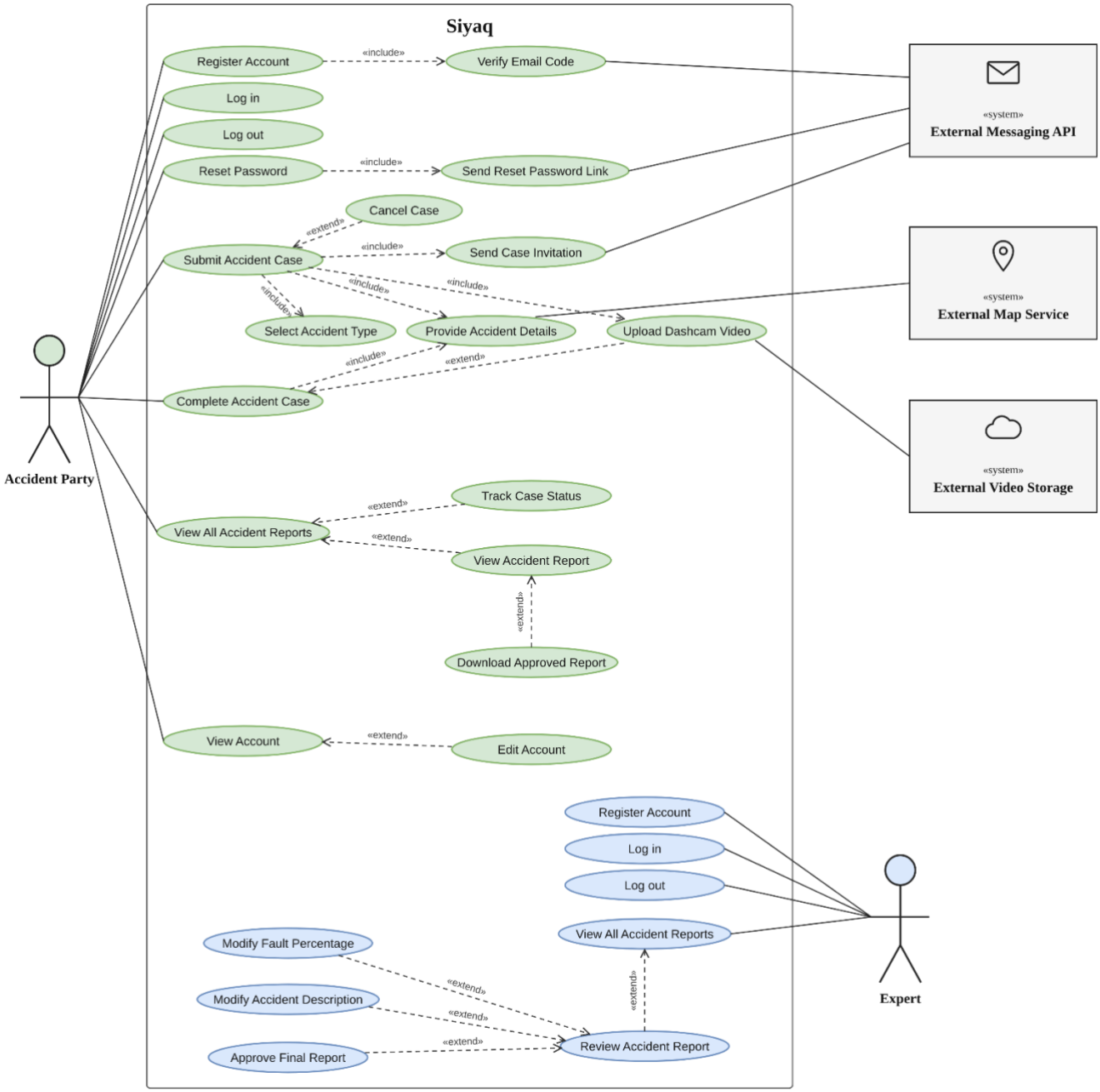


Figure 26: Use Case Diagram of SIYAQ.

6 Product Backlog

This section presents the Product Backlog of Siyaq, which lists the prioritized functional and non-functional requirements needed to deliver the system across its planned releases. Each entry in the backlog is a Product Backlog Item (PBI), written as a user story, and may represent a new feature, a defect, a technical improvement, or knowledge-acquisition work. Before a PBI is considered ready to be moved into a sprint, the team confirms it meets the criteria below.

Example Definition of Ready	
✓	Business value is clearly articulated
✓	Details are sufficiently understood
✓	Dependencies are identified; no blocking dependencies exist
✓	Team is appropriately staffed relative to the PBI
✓	Estimated and small enough to be completed during sprint
✓	Acceptance criteria are clear and testable
✓	Performance criteria, if any, are defined and testable
✓	Team understands how to demo the completed PBI

Table 16: Example Definition Ready Table.

6.1 Product Backlog Table

This section presents the team's product backlog, organized as user stories that describe Siyaq's features from the perspective of its users. Each entry includes the item's size, type, and acceptance criteria. The backlog covers all anticipated work needed to deliver the system, including new features, expert-review functionality, and non-functional requirements such as performance and usability.

PBI Numbers	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
	Account Access - Accident Party			
1	As an accident party, I want to create an account using my personal information and email address, so that I can report accidents and access my previous reports.	5	Feature	• The system requires the name, date of birth, national ID, mobile number, and a password. • The national ID is the identifier used to log in afterwards. • The email address is verified by a 6-digit code sent by email, and the account is activated only after the code is verified. • The system rejects registration when an account already exists for the same national ID, mobile number or email address. • The system displays a clear message for each invalid or missing field. • After successful registration the user is directed to the home screen.

PBI Numbers	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
2	As an accident party, I want to log in using my national ID and password, so that I can report a new accident or follow an existing case.	3	Feature	• The user logs in using the registered national ID and password. • Invalid credentials are rejected with a clear message that does not reveal which field was incorrect. • The account is temporarily locked after a defined number of failed attempts. • After a successful login the user is directed to the home screen. • Any open case belonging to the user is shown on the home screen after login.
3	As an accident party, I want to log out of my account, so that my accident information is not accessible to anyone using the same device afterwards.	2	Feature	• Log out is available from the settings page. • The session is terminated after logging out. • Protected pages cannot be reopened through the browser history or the back button after logging out. • Returning to the web application requires logging in again.
4	As an accident party, I want to reset my password using a secure reset link sent to my registered email address, so that I can regain access to my account and my previous reports.	5	Feature	• If the user requests a password reset by entering the national ID registered to the account, then a reset link is sent to the email address registered to that account. • The reset link expires after 30 minutes, after which it can no longer be used. • A new password can be set only by opening a valid, non-expired reset link. • The previous password is no longer accepted after the reset. • The system does not reveal whether the entered national ID belongs to a registered account.
5	As a second accident party, I want to log in or create an account when I open the case link, so that my information can be saved and reused for future cases.	5	Feature	• If the second accident party opens the case link and has no account, then they are prompted to register before proceeding. • If the second accident party opens the case link and already has an account, then they are prompted to log in before proceeding. • After successful login or registration, the second accident party is directed to the case form linked to that specific invitation link. • The link gives access only to the case it was issued for, and cannot be used to access any other case.
	Accident Case Submission - Accident Party			



PBI Numbers	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
6	As an accident party, I want to start a new report by tapping a single "Report Accident" button on my home screen, so that I can begin the process quickly without navigating through multiple menus.	2	Feature	• If the accident party is logged in, then the "Report Accident" button is displayed on the home screen. • If the accident party taps "Report Accident," then the case-submission form opens directly. • If the case-submission form opens successfully, then the accident party can begin entering the required accident information. • If the form cannot be opened due to a system error, then the system displays a clear error message.
7	As an accident party, I want my personal information to be pre-filled from my account and my vehicle details to be selected from my saved vehicles or entered manually, so that I don't have to repeatedly re-enter the same data.	6	Feature	• The accident party's name, date of birth, national ID, phone number and email are retrieved from the account and displayed read-only in the case form. • Personal information retrieved from the account cannot be edited within the case. • If the account has one or more saved vehicles, then the accident party can select the vehicle involved in this accident instead of re-entering its details. • If no vehicle is saved, or the vehicle involved is not among the saved ones, then the accident party enters the vehicle details manually. • If any required vehicle field is missing, then the system prevents submission and highlights the missing field.
8	As an accident party, I want the other party's personally identifying information to be masked in my case report -including their name, date of birth, national ID, phone number, and the vehicle owner's name- while the remaining vehicle details stay visible, so that their privacy is protected without limiting information needed to identify the vehicle.	8	Feature	• If the first accident party views the report, then the second party's name, date of birth, national ID, phone number, and vehicle owner name are masked. • If the second accident party views the report, then the first party's name, date of birth, national ID, phone number, and vehicle owner name are masked. • Vehicle details other than the owner's name (such as make, model, plate number) remain visible to both parties. • If an unauthorized user attempts to access another party's masked information, then access is denied.



PBI Numbers	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
9	As an accident party, I want to write a brief description of how the accident happened, so that the expert has additional context beyond the video evidence.	3	Feature	• If the accident party reaches the accident description step (step 4, after the vehicle information), then the system displays picture choices based on the accident type and automatically creates a suggested description from their selections. • If the accident is a rear-end collision, then the suggested description states whether the party's vehicle was in front or behind and what happened before the impact, based on their selections. • If the accident is a lane-change collision, then the suggested description states which side the other vehicle was on and which vehicle changed lanes, based on the party's selections. • If the party selects an impact area on the vehicle drawing, then the system includes that area in the suggested description. • If the party wants to change the suggested description, then they can edit it, add to it, or rewrite it entirely in their own words without being required to keep the suggested text. • If the description contains fewer than 10 characters or more than 500 characters, then the system prevents the party from continuing and displays a message stating that the description must contain between 10 and 500 characters.
10	As the first accident party, I want to upload a mandatory dashcam video when submitting my case, so that the system has visual evidence to analyze.	5	Feature	• If the first accident party does not attach a video, then the system prevents case submission and displays a message that the video is required. • If the uploaded file is not a supported video format, then the system rejects it and asks for a valid file.
11	As the first accident party, I want to provide the email address of the second accident party, so that they can be linked to the same case and asked to complete their portion.	3	Feature	• If the entered email address is invalid or is the first party's own email, then the system displays an error and prevents sending. • If the invitation is sent successfully, then the second accident party's email address is linked to the case.
12	As the first accident party, I want to submit the case after	5	Feature	• If all required fields (personal information, vehicle information, accident description, dashcam video and privacy consent) are



PBI Numbers	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
	completing all required information, so that the case is officially recorded and the second party can be invited to complete their part.			completed, then the "Submit" button becomes enabled. • If any required section is incomplete, then the "Submit" button remains disabled until all sections are completed. • The case is created with a status of "Pending" as soon as the accident information (Step 1) is saved, so the accident party can leave and resume it later. • If the case is submitted successfully, then an invitation link is sent to the second accident party's email address to complete their portion.
13 (new)	As the first accident party, I want to cancel a case before inviting the second party, so that I can discard a case created by mistake.	3	Feature	• A “Cancel case” option is available to the first accident party in every step before the invitation step. • If the accident party confirms the cancellation, then the case and all its entered information are deleted and no longer appear in their case list or the expert’s list. • Once the first accident party reaches the invitation step, the cancel option is no longer available and they must enter the second party’s email address to continue.
14	As an accident party, I want to view the current status of my case (Pending, In Progress, or Completed), so that I know where my report stands.	3	Feature	• If the second accident party has not yet submitted their portion, then the case status is displayed as "Pending." • If the second accident party has submitted and the case is awaiting expert review, then the case status is displayed as "In Progress." • If the expert has approved the case, then the case status is displayed as "Completed." • If the accident party opens their profile page, then a list of all their submitted cases is displayed.
15	As an accident party, I want to receive notifications when my case status changes and when the final result is approved, and to turn them on or off from my settings, so that I can follow my case without checking it repeatedly.	3	Feature	• If the case status changes, a notification is sent to the party's registered email address. • If the expert approves the final report, both parties are notified that the final result is available. • The party can enable or disable each notification type from the profile settings. • If a notification type is disabled, no notification of that type is sent.



PBI Numbers	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
16	As an accident party, I want to view my personal information, change my email address, and add, edit, or remove my saved vehicles through my profile settings, so that my account information stays accurate and my vehicles are ready for future cases.	5	Feature	• If the accident party opens their profile settings, then they can view their saved personal information and vehicles. • The accident party's full name, national ID/Iqama number, date of birth, and mobile number are displayed as read-only fields. • If the accident party requests an email change, then a verification code is sent to the new email address. The email is updated only after successful verification. • If the new email address is invalid or registered to another account, then the system displays an error and keeps the current email unchanged. • If the accident party adds a vehicle with valid manufacturer, model, manufacturing year, plate details, and color, then it is saved to their account and becomes available for selection in future cases. • If the accident party edits a saved vehicle, then the updated details are saved and displayed when selecting that vehicle for future cases. • If the vehicle details are invalid or its plate is already registered to another vehicle in the same account, then the system displays an error and does not save the changes. • If the accident party confirms the removal of a saved vehicle, then it is removed from their account and is no longer available for selection in future cases.
17	As an accident party, I want to download my case's final report as a PDF file, so that I can keep a copy for my own records.	5	Feature	• If the case status is "Completed", then the accident party can download the final report as a PDF. • The downloaded PDF includes the same masked/visible information rules as the in-app report (personal information of the other party remains masked). • If the case is not yet "Completed", then the download option is not available.
	Account Access - Authorized Expert			
18	As an authorized expert, I want my registration to be accepted only when I provide a valid authorization code, so that no unauthorized person	5	Feature	• Registration first requires the expert's full name and the authorization code provided to authorized experts. • If the entered authorization code does not match the code configured in the system, then registration is rejected with a clear message.



PBI Numbers	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
	can reach the expert interface.			• After the code is verified, the expert chooses an expert identifier (4 to 32 English letters or digits) and a password of at least 12 characters, confirmed twice. • An expert identifier that is already used by another account is rejected. • Only an authenticated expert account can open the expert interface.
19	As an authorized expert, I want to log in using my expert identifier and password, so that I can review the accident cases waiting for me.	3	Feature	• The expert logs in using the expert identifier chosen at registration and a password. • After a successful login the expert is directed to the list of cases rather than the accident-party home screen. • Invalid credentials are rejected with a clear message that does not reveal which field was incorrect. • The account is temporarily locked after 5 failed attempts for 15 minutes. • Each login is recorded with the expert identifier and the time.
20	As an authorized expert, I want to log out of the expert interface, so that no accident case is left accessible on an unattended device.	2	Feature	• Log out is available from any page of the expert interface. • Any case opened for review is released so that it is not left locked for other reviewers. • The session is terminated and case data cannot be restored through the browser history. • Returning to the expert interface requires logging in again.
	Case Completion & Expert Review			
21	As a second accident party, I want to access the accident case through the received link and, after logging in, complete my required information, so that I can provide my details for the accident assessment.	5	Feature	• If the second accident party opens the received link and successfully logs in or registers, then the accident case form should be displayed. • If the second accident party enters all required information correctly, then the system should allow them to continue. • If required information is missing, then the system should indicate the missing fields and prevent submission.
22	As a second accident party, I want to view the basic accident details through the	2	Feature	• If the second accident party opens the received link, then the case's accident date, accident type, and the first party's vehicle make, model, and plate number are displayed.



PBI Numbers	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
	received link, so that I can confirm the correct case.			• The first party's personal information is not displayed.
23	As a second accident party, I want to upload my dashcam video if I have one, so that the system can use it as additional evidence for the accident analysis.	3	Feature	• If the second accident party has a dashcam video, then the system should allow them to upload the video. • If the uploaded video is in a supported format, then the system should accept the video. • If the uploaded video is invalid or unsupported, then the system should display an error message and allow the user to upload another video. • If the second accident party does not have a dashcam video, then they should be able to continue without uploading one.
24	As a second accident party, I want to submit my completed accident information, so that the system can start analyzing the accident case.	3	Feature	• If the second accident party has completed all required information, then the system should allow them to submit the accident case. • If the accident case is submitted successfully, then the system should confirm that the case has been submitted. • If required information is missing, then the system should prevent the submission and indicate what needs to be completed. • Once both parties have submitted and the case status is "In Progress", neither party can edit the case data.
25	As an authorized expert, I want my case list organized into three sections by status (Pending, In Progress, Completed), so that I can easily see which cases are awaiting the second party, which are ready for my review, and which I've already completed.	5	Feature	• If the second accident party has not yet submitted their portion, then the case appears under the "Pending" section. • If the second accident party has submitted and the analysis is complete, then the case appears under the "In Progress" section, ready for the expert's review. • If the expert has approved the case, then it appears under the "Completed" section.
26	As an authorized expert, I want to view the generated report and its supporting accident frames, so that I can	5	Feature	• If the accident analysis is completed, then the authorized expert should be able to view the generated final report. • If the expert opens the case, then the supporting accident frames should be available as attachments.

PBI Numbers	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
	review the system's analysis and evidence.			• If the report and supporting frames are available, then the expert should be able to review them before making a decision.
27 (new)	As an authorized expert, I want to view the fault rules used by Siyaq, so that I can check the suggested fault percentage for a case.	2	Feature	• A "Rules" page is available in the expert interface next to "Cases". • The page lists the rear-end rules (R1–R3) and the lane-change rules (L1–L6), with the situation, required evidence, violated provision and suggested fault ratio for each rule. • The page explains how fault ratios are assigned (100/0, 75/25, 50/50). • Only logged-in experts can open the page.
28	As an authorized expert, I want to approve the generated report, so that it can be finalized and sent to the accident parties.	3	Feature	• If the authorized expert has reviewed the final report and accepts the result, then the expert should be able to approve the report. • If the report is approved, then it should be ready to be sent to the accident parties.
29	As an authorized expert, I want to modify the fault percentage or accident description, so that I can adjust the report before it is finalized.	5	Feature	• If the authorized expert identifies an issue in the fault percentage or accident description, then the expert should be able to modify it. • If the expert modifies the fault percentage or accident description, then the updated information should be reflected in the final report. • If the expert completes the required modifications, then the expert should be able to finalize the report.
	User Interface and Interaction – Accident Party			
30	As an accident party, I want to view my home interface after logging in, so that I can easily access my accident cases and the functions available to me.	5	Feature	• The accident party is directed to the accident party home interface after successful login. • The interface displays the main functions available to the accident party. • The user can access the option to start a new accident case. • The user can access their existing accident cases. • Each existing case displays its current status. • Only cases associated with the logged-in accident party are displayed. • If no accident cases exist, an appropriate message is displayed.

PBI Numbers	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
31	As an accident party, I want to view my accident cases in one place, so that I can easily identify, access, and follow each case.	3	Feature	• All accident cases associated with the logged-in user are displayed. • Each case displays sufficient information to distinguish it from other cases. • The current status of each case is displayed. • The user can select a case to view its available information. • Cases belonging to other users are not displayed. • The displayed information corresponds to the selected case.
32	As an accident party, I want to view the final approved accident assessment, so that I can understand the final result of my case.	5	Feature	• The final assessment is available only after expert approval. • The final fault percentage is clearly displayed. • The assessment is identified as the final approved result. • Available supporting information is presented with the result. • An unapproved system-generated suggestion is not presented as the final result. • The displayed result belongs to the selected accident case.
	Non-Functional Requirements			
33	As an accident party, I want the complete accident evaluation process, from submitting the accident case and uploading the video until the expert receives the generated report, to be completed within 10 minutes, so that I can receive the final result in a shorter time.	8	Feature	• The 10-minute timer starts when the accident party submits the case and video. • The system processes the uploaded video and completes the accident analysis within the defined time. • The generated report is available to the expert within 10 minutes of case submission.
34	As an accident party, I want the system to be available when I need to submit or review an accident case, so that I can access the service without unnecessary interruptions.	5	Feature	• The system is available 99% of the time. • Users can submit accident cases at any time. • Users can access their submitted cases and reports whenever the system is available.

PBI Numbers	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
35	As an accident party, I want the system to have a simple and clear interface, so that I can submit my accident information and dashcam video without needing advanced technical skills.	5	Feature	• The system provides a clear and consistent Arabic interface. • The steps required to submit an accident case are clearly displayed. • Required fields and actions are clearly identified. • Users can upload a dashcam video without needing advanced technical knowledge. • The system provides clear instructions and error messages when the user enters invalid or missing information.
36	As an accident party, I want the system to clearly handle unsupported or invalid video files, so that I know what went wrong and how to continue.	3	Feature	• The system validates the uploaded video before starting the analysis. • Unsupported video formats are rejected. • Invalid or corrupted video files are detected and rejected. • The system displays a clear error message explaining why the video cannot be processed. • The user can remove the invalid file and upload another supported video without restarting the entire case submission process.
37	As an accident party, I want my uploaded video and personal information to be handled securely, so that my privacy is protected.	8	Feature	• Only authorized users can access the submitted accident information and video. • The personal information of the other accident party is not fully displayed to unauthorized users. • The system protects uploaded videos and personal information from unauthorized access. • Uploaded videos and associated personal data are automatically deleted after the defined retention period.

Table 17: Product Backlog.

7 Appendices

Appendix A: Project Management Tools

Jira link: Click here

Github link: Click here

Appendix B: Questionnaire Study For Requirements Elicitation

Questionnaire Form : Click Here

Questionnaire Questions :

1. الفئة العمرية

   - 18 - 24
   - 25 - 34
   - 35 - 44
   - 45 اكثر من

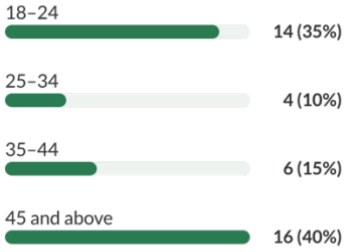


Figure 27: Age Distribution of Participants.

2. الجنس

   - أنثى
   - ذكر



Figure 28: Gender Distribution of Participants.

3. منذ كم سنة وأنت تقود السيارة؟
   - أقل من سنة
   - من سنة إلى ثلاث سنوات
   - من اربع سنوات إلى سبعة سنوات
   - أكثر من سبعة سنوات

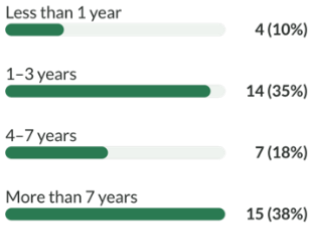


Figure 29: Participants' Years of Driving Experience.

4. ما مدى خبرتك في استخدام تطبيقات الويب؟

- مبتدئ، أستخدم مواقع الويب الأساسية فقط وأحتاج للمساعدة أحيانًا في رفع الملفات أو إكمال بعض الإجراءات
- متوسط، أستخدم مواقع الويب بشكل منتظم وأستطيع رفع الملفات وإكمال معظم الإجراءات دون مساعدة
- متقدم، أستخدم تطبيقات الويب باستمرار وأجيد التعامل مع مختلف الميزات وحل المشكلات البسيطة

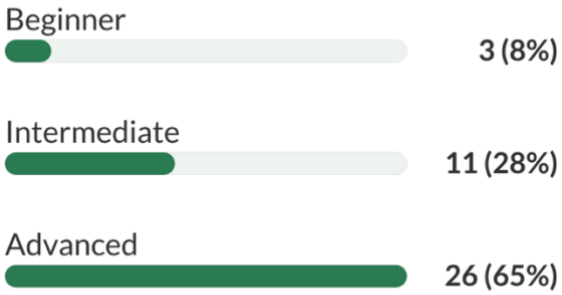


Figure 30: Participants' Experience Level with Web Applications.

5. هل سبق لك تنزيل أو نقل مقاطع من كاميرا القيادة (داشكام) إلى الهاتف أو الكمبيوتر؟

- نعم، بسهولة
- لا، لكن اعرف الطريقة
- لا، ولا اعرف الطريقة

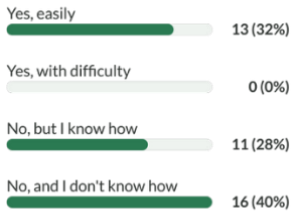


Figure 31: Prior Experience Transferring Dashcam Footage.

6. هل تستخدم كاميرا قيادة (داشكام) في مركبتك؟

- نعم
- لا



Figure 32: Dashcam Usage Among Participants.

7. إذا كانت الإجابة لا، هل تفكر باستخدامها مستقبلاً؟

- نعم
- ربما
- لا

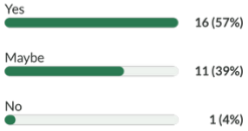


Figure 33: Willingness to Adopt a Dashcam in the Future.

8. هل سبق لك التعرض لحادث مروري؟
   - نعم
   - لا



Figure 34: Prior Exposure to a Traffic Accident.

9. كيف تم تقييم الحادث؟
   - حضور مختص إلى الموقع
   - عن بعد
   - أخرى

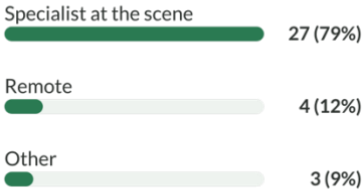


Figure 35: Method Used for the Accident Assessment.

10. كم استغرقت عملية تقييم الحادث تقريبًا؟
    - اقل من ساعة
    - من ساعة الى 3 ساعات
    - من 3 ساعات الى يوم



Figure 36: Duration of the Accident Assessment

11. ما مدى رضاك عن آلية تقييم الحادث؟
    - غير راضٍ
    - محايد
    - راضٍ
    - راضٍ جدًا

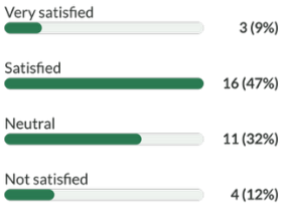


Figure 37: Participants' Satisfaction with the Assessment Process.

12. في حال كنت راضٍ عن آلية التقييم، ما سبب رضاك؟

- سهولة الإجراءات
- سرعة إنهاء الطلب
- وضوح النتيجة
- تعامل الموظفين
- إمكانية تقديم الأدلة
- أخرى

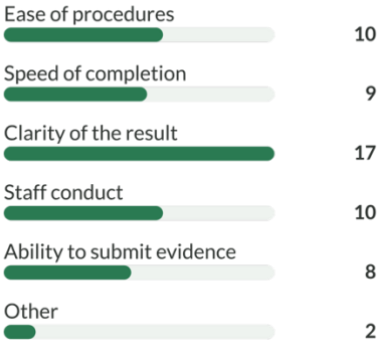


Figure 38: Reasons for Satisfaction with the Assessment Process.

13. في حال لم تكن راضٍ عن تقييم الحادث، ما سبب عدم رضاك؟

- طول مدة الانتظار
- عدم وضوح سبب تحديد نسبة الخطأ
- صعوبة تقديم الأدلة
- تأخر ظهور النتيجة
- أخرى

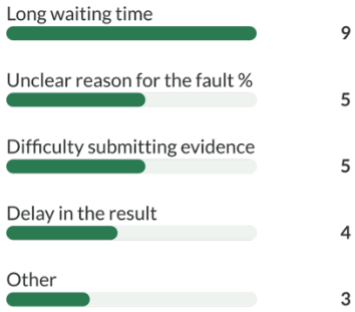


Figure 39: Reasons for Dissatisfaction with the Assessment Process.

14. برأيك، إلى أي مدى تساعد تسجيلات كاميرا القيادة (الداشكام) في توضيح ملابسات الحوادث المرورية؟

- تساعد جدًا
- تساعد
- محايد
- لا تساعد

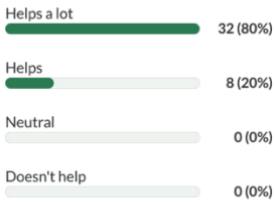


Figure 40: Perceived Usefulness of Dashcam Footage in Clarifying Accidents.

15. تم اقتراح نظام جديد باسم "سياق" يعتمد على الذكاء الاصطناعي لتحليل مقاطع كاميرات القيادة (الداشكام) للحوادث المرورية البسيطة، بهدف تسريع عملية تقييم الحادث من خلال تحليل الفيديو وإظهار الأدلة التي استند عليها التحليل، وفي حال توفر النظام، إلى أي مدى تؤيد فكرة النظام؟

- أؤيده بشدة
- أؤيده
- محايد
- لا أؤيده

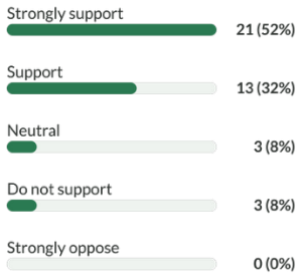


Figure 41: Level of Support for the Proposed Siyaq System.

16. في حال توفر هذا النظام، هل ستفضل استخدامه عند التعرض لحادث مروري بسيط؟

- نعم
- ربما
- لا

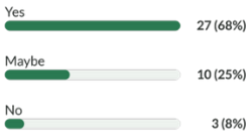


Figure 42: Willingness to Use Siyaq for a Simple Traffic Accident.

17. ما أهم الميزات التي تود توفرها في النظام؟ (يمكن اختيار أكثر من إجابة)

- سرعة تحليل الحادث
- عرض الأدلة التي بُني عليها التحليل
- سهولة رفع الفيديو
- واجهة سهلة الاستخدام
- إمكانية متابعة حالة الطلب
- أخرى

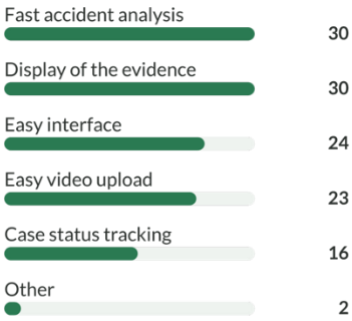


Figure 43: Most Desired Features in the Siyaq System.

18. ما الميزات أو الخدمات الأخرى التي تتمنى إضافتها في نظام "سياق"؟ (إجابة مفتوحة)

- يكون تحليل الذكاء الاصطناعي كخيار وليس بالاجبار، وذلك لاعتماد حرية الشخص بانتظار وصول شخص مختص يحلل الحادث او رغبتهم بانهاء العملية بسرعة وذلك باستخدام الاداء.. فيحبذ أن يكون أداة مساعدة وليست أساسية
- إضافة تنبيهات للحالات المهمة وإمكانية حفظ التقارير والرجوع لها لاحقًا
- عدم إمكانية جهات أخرى من الحصول على التقرير

- دقة تحديد نسبة الخطأ
- امكانية الاعتراض وطلب النظر في التقييم من اشخاص طبيعيين في حالة الاعتراض



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